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LITERATURE REVIEW

Paper 1:

ADPM: An Alzheimer's Disease Prediction Model for Time Series Neuroimage Analysis (2020)

Authors: Xin Hong; Rongjie Lin; Chenhui Yang; Chunting Cai; Kathy Clawson.

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is a form of dementia which causes memory, thinking, and behavior disorders in humans. Effective early diagnosis and treatment of AD is of fundamental importance as it can reduce disease progression, allow more effective management of symptoms, facilitate timely patient access to advice and support, and lower associated costs of health care. Given that Alzheimer's typically progresses in stages over an extended period of time, we propose that automated analysis of time sequential data may enhance disease prediction. We present a novel time-series Alzheimer's Disease Prediction Model (ADPM) comprising Random Forest (RF) region of interest (ROI) selection and Gated Recurrent Units (GRU) prediction. Experiments show that our methodology achieves higher classification accuracy in comparison to existing algorithms, and can facilitate prediction of early onset AD. Furthermore, testing demonstrates that random forest ROI selection can identify disease-relative brain regions across different image modalities (MRI, PET, DTI).

Paper 2:

Improving Alzheimer's Detection With Deep Learning and Image Processing (2024)

Authors: Ghadah Naif Alwakid; Sidra Tahir; Mamoona Humayun; Walaa Gouda

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is characterized by the gradual degeneration and decline of brain cells, leading to irreversible neurological changes. This study investigates advanced image enhancement techniques for improving AD diagnosis using brain MRI. The methods used include CLAHE (Contrast Limited Adaptive Histogram Equalization) to enhance local image contrast and ESRGAN (Enhanced Super-Resolution Generative Adversarial Networks) to improve image resolution. These preprocessing methods improve MRI images and classification accuracy. An ensemble model of MobileNetV2 and DenseNet121, two efficient deep-learning models with feature extraction capabilities, were used as classifiers. This approach addresses challenges in AD diagnosis by leveraging deep learning for more accurate classification of brain tissue images. The model achieved an accuracy of 80.31% for MobileNetV2 and 89.22% for DenseNet121 when no enhancements were utilized. The model achieved accuracies of 92.34% and 89.38% for MobileNetV2 and DenseNet121, respectively, when enhancement methods were utilized, indicating strong dependability with the Kaggle dataset of MRI images. These findings underscore the efficacy of advanced image processing and deep learning in the early and accurate detection of Alzheimer's disease.

Paper 3:**Dual Attention Convolutional AutoEncoder for Diagnosis of Alzheimer's Disorder in Patients Using Neuroimaging and MRI Features(2024)**

Authors: Tanzila Saba; Shaha Alotaibi; Muhammad Mujahid; Amjad Rehman; Haitham Nobanee; Javed Ahmad

Journal: IEEE Access

Abstract:

Alzheimer's disease is a neurodegenerative disease causing memory loss and brain protein accumulation. Early diagnosis is crucial for clinical trials and patient care. Magnetic resonance imaging (MRI) methods have improved diagnosis and prognosis, but doctors need to interpret images proficiently. Deep learning technology has shown potential in detecting Alzheimer's disease, but the disease progresses slower in early phases. A new dual-attention convolutional autoencoder model is presented, offering improved detection abilities and potential for real-time use in Alzheimer's disease

diagnosis. The study utilized two datasets: the first ADNI dataset, which includes three classes (MCI, CN, and AD), and the second Alzheimer's Disease Neuroimaging Dataset, which includes two distinct classes (AD and MCI). We analyze the effectiveness of our proposed model by evaluating key performance metrics such as accuracy, precision, sensitivity, specificity, F1 score, and AUC score. In addition, we utilize cross-validation and mean absolute error to validate our model while also fine-tuning the parameters. Based on experimental data, the proposed model accurately detected Alzheimer's disease with an accuracy of 0.9902 ± 0.0139 . Based on the results, the proposed model demonstrates excellent performance compared to the existing methods described in the literature. The proposed model achieves precision, sensitivity, and specificity of 0.9882 ± 0.0587 , 0.9898 ± 0.0865 , 0.9912 ± 0.0872 respectively. The model achieved an AUC score of 0.9992 for MCI and 0.9919 for AD class. Furthermore, the proposed method can enhance the affordability of Alzheimer's disease diagnostics and increase the rate of early AD detection by facilitating remote healthcare.

Paper 4:

Graph Wavelet-Based Multilevel Graph Coarsening and Its Application in Graph-CNN for Alzheimer's Disease Detection (2024)

Authors: Karan Kumar; Isha Suwalka; Adaora Uche-Ezennia; Celestine Iwendi; Cresantus N. Biamba.

Journal: IEEE Access

Abstract:

Along with the classical applications like graph partitioning, graph visualization, etc., graph coarsening has been recently applied in graph convolutional neural network (GCNN) architectures to perform the pooling operation in the graph domain. In this paper, we propose a novel two-stage graph coarsening method rooted on the graph signal processing with its application in the GCNN architecture. In the first stage of coarsening, the graph wavelet transform (GWT) based features are used to obtain a coarsened graph which preserves the topological characteristics of the original graph. In the second stage, the coarsening problem is formulated as an optimization problem where the reduced Laplacian operator at each level is obtained as a restriction of the original Laplacian operator to a specified subspace that also maximizes the topological similarity. The performance of the proposed coarsening algorithm is quantified in the

general coarsening context using different graph coarsening quality measures. Its effectiveness as a pooling operator in GCNN is validated by applying it for the graph coarsening operation in the GCNN architecture. This modified GCNN architecture is then used as a graph signal classifier for the early detection of Alzheimer's disease. The results show that the proposed coarsening method outperforms state-of-the-art methods, both in the general coarsening context and as a pooling operator in the GCNN architecture.

Paper 5:

A Deep Convolutional Neural Networks Based Approach for Alzheimer's Disease and Mild Cognitive Impairment Classification Using Brain Images (2022)

Authors: Ruhul Amin Hazarika; Debdatta Kandar; Arnab Kumar Maji.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a hazardous neurological disorder of people aged in the early 60s. The main symptoms of AD is significant memory loss. Mild Cognitive Impairment (MCI) is a state of dementia in which a patient exhibits the early symptoms of AD. Since brain is the most impacted region, the disorders can be classified by analyzing factors from brain tissues in different subjects. Machine Learning (ML) is a widely utilised concept that aids in the decision-making process. Deep Convolutional Neural Network (DNN) is a type of ML techniques that uses artificially connected neurons to mimic the human brain. In this work, we have proposed a novel DNN-based model for distinguishing AD and MCI patients from Cognitively Normal individuals. Inspired by the original VGG-19, we have created 19 deep layers in the network. In Back Propagation, deeper models suffer from the problem of vanishing gradient and information loss. As a solution, we borrowed the Dense-Block notion from the original DenseNet architecture, which provides a path of information exchange amongst all the layers. Furthermore, we have implemented depth-wise convolutional procedures to make the model computationally faster. Outcome of the proposed model is compared with some prominent DNN models and observed that, the proposed approach performs most convincingly with an average performance rate of 95.39%.

Paper 6:

TLEABLCNN: Brain and Alzheimer's Disease Detection Using Attention Based Explainable Deep Learning and SMOTE Using Imbalanced Brain MRI (2025)

Authors:Erol Kina.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is one of the primary causes of dementia. It degenerates the brain and reduces the activity of individuals by disrupting their memory and physiological functions. A comprehensive examination of specific brain tissue is required in order to accurately diagnose a brain condition using magnetic resonance imaging. The main aim of this research was to develop a rapid and effective technique for detecting healthy persons before the onset of brain tumours, including AD, pituitary tumours, gliomas, and meningiomas. This work presents a lightweight convolutional architecture based on EfficientNet with a Squeeze Attention Block using transfer learning. The proposed approach used lightweight layers with an L2 regularizer, global pooling (2D), and batch normalisation to construct the model, including two dropout layers. This paper utilises the synthetic minority oversampling approach to address the issues of overfitting and imbalanced samples by balancing two significantly imbalanced MRI datasets. We evaluated the efficacy of the minority strategy by doing experiments on both the training set and the entire dataset. The proposed model demonstrates high efficiency, with the risk of overfitting mitigated by applying SMOTE solely to the training data, while the test and validation datasets remain unaffected. The effectiveness of the proposed approach was evaluated using several performance measures and compared with previous published research. We evaluated the proposed approach for explainability via gradient-weighted class activation mapping to comprehend the model's behaviour and its predictions. The proposed framework offers advantages over existing models in terms of computing efficiency, explainability, generalization, and clinical significance.

Paper 7

A Modality-Flexible Framework for Alzheimer's Disease Diagnosis Following Clinical Routine(2024)

Authors:Yuanwang Zhang; Kaicong Sun; Yuxiao Liu; Fang Xie

Journal: IEEE Access

Abstract:

Dementia has high incidence among the elderly, and Alzheimer's disease (AD) is the most common dementia. The procedure of AD diagnosis in clinics usually follows a standard routine consisting of different phases, from acquiring non-imaging tabular data in the screening phase to MR imaging and ultimately to PET imaging. Most of the existing AD diagnosis studies are dedicated to a specific phase using either single or multi-modal data. In this paper, we introduce a modality-flexible classification framework, which is applicable for different AD diagnosis phases following the clinical routine. Specifically, our framework consists of three branches corresponding to three diagnosis phases: 1) a tabular branch using only tabular data for screening phase, 2) an MRI branch using both MRI and tabular data for uncertain cases in screening phase, and 3) ultimately a PET branch for the challenging cases using all the modalities including PET, MRI, and tabular data. To achieve effective fusion of imaging and non-imaging modalities, we introduce an image-tabular transformer block to adaptively scale and shift the image and tabular features according to modality importance determined by the network. The proposed framework is extensively validated on four cohorts containing 6495 subjects. Experiments demonstrate that our framework achieves superior diagnostic performance than the other representative methods across various AD diagnosis tasks, and shows promising performance for all the diagnosis phases, which exhibits great potential for clinical application

Paper 8

A Review on Machine Learning Approaches for Diagnosis of Alzheimer's Disease and Mild Cognitive Impairment Based on Brain MRI (2024)

Authors: Helia Givian; Jean-Paul Calbimonte.

Journal: IEEE Access

Abstract:

Alzheimer's disease is a progressive disease for which researchers have yet to discover the main cause, but believe it probably involves a combination of age-related changes in the brain, genetic, environmental and lifestyle factors. Alzheimer's is an irreversible disease that still has no cure. Therefore, early diagnosis of that is very important to prevent its progression. Developing machine learning algorithms in health care, especially in brain disorders such as Alzheimer's disease provides new opportunities for early diagnosis and recognition of important biomarkers. This paper presents an overview of advanced studies based on machine learning techniques for diagnosing Alzheimer's disease and different stages of mild cognitive impairment based on magnetic resonance imaging images in the last 10 years. Also, this paper comprehensively describes the commonly efficient machine learning algorithms in each stage of magnetic resonance imaging processing used in the papers, which can facilitate the comparison of algorithms with each other and provide insight into the impact of each technique on classification performance. This review can be a valuable resource to gain a new perspective on the various research methods used in recent studies on Alzheimer's disease.

Paper 9:**Accelerating Training of Convolutional Neural Networks With Hessian-Free Optimization for Detecting Alzheimer's Disease in Brain MRI(2024)**

Authors: Marios Pafitis; Michalis Constantinou; Christos Christodoulou.

Journal: IEEE Access

Abstract:

Convolutional neural network (CNN) classifiers, which perform feature extraction from brain magnetic resonance imaging (MRI) data and classify them as healthy or diseased, are very promising in aiding the diagnosis of neurological disorders. CNN models are usually optimized with first-order algorithms, e.g. stochastic gradient descent (SGD) and adaptive momentum (Adam), which are susceptible to pathological curvature of the objective function. Hessian-free optimization (HFO), a second-order algorithm, overcomes this problem and thus is expected to converge faster. Here, we investigated whether HFO can increase the training speed of CNN models for brain MRI classification into cognitively normal or Alzheimer's disease using data from the

Alzheimer's Disease Neuroimaging Initiative. We trained two naive CNN classifiers with either HFO, SGD or Adam and compared their performance. Our results show that HFO converged at least two to four times faster than Adam and SGD. The cross-validation accuracy of the models optimized with HFO was comparable or better than those optimized with SGD or Adam. In addition, HFO was more robust to hyperparameter changes. These results comply with theory and act as a proof of concept that second-order algorithms have an advantage over first-order methods in training CNN models for MRI classification.

Paper 10:

AttCORAL: Domain-Adaptive Attention Networks for Early Alzheimer's Disease Diagnosis (2025)

Authors: Gia Minh Hoang; Nhat Hoang; Tran Thi Hoai Thu; Nghia Tien Trong Hoang; Jae Gwan Kim.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is one of the most common neurodegenerative disorders characterized by the progressive accumulation of amyloid-beta plaques and tau protein tangles in the brain. Due to the lack of a cure for Alzheimer's disease, early and accurate diagnosis of AD is crucial for effective early interventions to slow disease progression. Magnetic Resonance Imaging (MRI) has emerged as a promising modality for early diagnosis, providing detailed insights into brain structure alterations associated with AD. However, domain shift due to variations in imaging protocols and data distribution among national cohorts remains a challenge for the application of MRI in clinical diagnosis. To address this issue, we propose AttCORAL, a novel Domain-Adaptive Attention Network for Early Alzheimer's Disease Diagnosis, integrating attention mechanisms with Correlation Alignment (CORAL) loss to effectively mitigate domain discrepancy, enhancing the model's robustness and generalization. We evaluate AttCORAL on two large-scale MRI datasets—Alzheimer's Disease Neuroimaging Initiative (ADNI) and the Australian Imaging, Biomarkers, and Lifestyle Study (AIBL)—which differ in acquisition protocols and demographics. While these datasets provide a valuable basis for cross-cohort validation, we acknowledge that further multi-cohort studies are necessary to fully assess global generalizability. To ensure the reliability of our approach, we apply Grad-CAM to visualize the pathological brain regions most informative for our model's predictions. Experimental results

demonstrate that AttCORAL significantly outperforms current state-of-the-art studies, highlighting its effectiveness in early diagnosis of AD across diverse imaging domains.

Paper 11:

A Lightweight Deep Convolutional Neural Network Extracting Local and Global Contextual Features for the Classification of Alzheimer's Disease Using Structural MRI

Authors: Emimal Jabason; M. Omair Ahmad; M. N. S. Swamy.

Journal: IEEE Access

Abstract:

Recent advancements in the classification of Alzheimer's disease have leveraged the automatic feature generation capability of convolutional neural networks (CNNs) using neuroimaging biomarkers. However, most of the existing CNN-based methods often disregard the local features of the brain data, which leads to a loss of subtle fine-grained features in the brain imaging data. Moreover, the existing CNN architectures, which mainly rely on global features, do not pay much attention to the discriminability of the extracted features for the task of classification of Alzheimer's disease. Moreover, the existing architectures often end up using a large number of parameters to enhance the richness of the extracted features. This paper proposes a novel lightweight deep CNN, which extracts local and global contextual features from the sagittal slices of structural MRI data and uses both of these two types of features for the classification of the disease. The main idea used in designing the proposed network is to process separately the local and global features by using modules that pay a special attention to extract local and global contextual features. The fused local and global contextual features are then used for the classification of Alzheimer's disease. The proposed network is tested for the binary and multiclass classifications of the disease using the MR images taken from the ADNI database. The proposed network is shown to provide a performance that is significantly higher than that provided by other existing state-of-the-art networks, yet using a number of parameters that is a small fraction of that used by the other schemes.

Paper 12:

A Novel CNN Architecture for Accurate Early Detection and Classification of Alzheimer's Disease Using MRI Data(2024)

Authors: . A. M. El-Assy; Hanan M. Amer; H. M. Ibrahim; M. A. Mohamed.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a debilitating neurodegenerative disorder that requires accurate diagnosis for effective management and treatment. In this article, we propose an architecture for a convolutional neural network (CNN) that utilizes magnetic resonance imaging (MRI) data from the Alzheimer's disease Neuroimaging Initiative (ADNI) dataset to categorize AD. The network employs two separate CNN models, each with distinct filter sizes and pooling layers, which are concatenated in a classification layer. The multi-class problem is addressed across three, four, and five categories. The proposed CNN architecture achieves exceptional accuracies of 99.43%, 99.57%, and 99.13%, respectively. These high accuracies demonstrate the efficacy of the network in capturing and discerning relevant features from MRI images, enabling precise classification of AD subtypes and stages. The network architecture leverages the hierarchical nature of convolutional layers, pooling layers, and fully connected layers to extract both local and global patterns from the data, facilitating accurate discrimination between different AD categories. Accurate classification of AD carries significant clinical implications, including early detection, personalized treatment planning, disease monitoring, and prognostic assessment. The reported accuracy underscores the potential of the proposed CNN architecture to assist medical professionals and researchers in making precise and informed judgments regarding AD patients.

Paper 13:

Early Diagnosis of Alzheimer's Disease Using Adaptive Neuro K-Means Clustering Technique (2025)

Author:. Bijen Khagi; Goo-Rak Kwon.

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder characterized by memory loss, behavioral changes, and impaired self-care, often preceded by Mild Cognitive Impairment (MCI). Not all MCI cases progress to AD, creating a diagnostic

challenge. This study proposes a novel framework for early AD diagnosis using T1-weighted Magnetic Resonance Imaging (MRI). The approach integrates the Adaptive Moving Self-Organizing Map (AMSOM), a neural network technique for unsupervised training and tissue segmentation, with K-means clustering and Principal Component Analysis (PCA) for feature selection. AMSOM dynamically updates neuron weights to improve segmentation accuracy. Classification is performed using various algorithms, evaluated on sensitivity, accuracy, precision, and similarity metrics. Compared to existing techniques such as Fuzzy C-means (FCM) and hybrid Self-Organizing Mapping-K-means (SOM-FKM), the proposed method demonstrates statistically significant improvements in tissue segmentation and classification. It achieved a mean accuracy of 99.8%, reducing the Mean Squared Error (MSE) from 2.3 to 0.44 and improving the Discriminative Overlap Index (DOI) and Tissue Clarity (TC) values to 0.435105 and 0.282381, respectively. Implemented in MATLAB, this method provides a robust, efficient framework for early AD detection, surpassing existing approaches in precision and reliability.

Paper 14:

A Review on Alzheimer's Disease Through Analysis of MRI Images Using Deep Learning Techniques (2023)

Author:Yonglin Chen et al.

Journal: IEEE Access

Abstract

The anatomical structure of the brain has been studied with the help of magnetic resonance imaging (MRI), which has been used to analyze numerous neurological diseases and define pathological areas. Early detection of Alzheimer's Disease (AD) patients is critical in order to implement preventative measures. Alzheimer's disease (AD) is the most common chronic disease in the elderly, with a high incidence rate. In recent years, deep learning has seen a lot of success in the medical image analysis. Brain diseases can be more accurately categorized using segmented MRI scans due to in-depth analyses of tissue architecture. Many, complex segmentation approaches have been presented for AD diagnosis. Since deep learning algorithms can yield effective results over a large data collection, they have received interest for use in segmenting the brain's structure and classifying AD. Consequently, the deep learning techniques are currently favored over machine learning techniques. We discuss how convolutional neural network concepts can be used to study brain anatomy in order to detect AD. New

techniques, their results on open datasets, and the benefits of brain MRI segmentation for Alzheimer's disease categorization are discussed. In this article, the literature on Alzheimer's disease is briefly reviewed, and the possibility of Deep Learning to improve early diagnosis is discussed.

Paper 15:

Same-Subject-Modalities-Interactions: A Novel Framework for MRI and PET Multi-Modality Fusion for Alzheimer's Disease Classification (2023)

Author:Guelib, B., Zarour, K., Hermessi, H., Bounab, R., & Khelifa, N. (2023)

Journal: IEEE Access

Abstract:

Alzheimer's disease is a growing concern, and neuroimaging techniques such as Magnetic Resonance Imaging (MRI) and Positron Emission Tomography (PET) scans are widely used to classify AD patients. While MRI captures structural information and measures brain atrophy, PET shows functional changes associated with neurological disorders, and both modalities have been proven to be AD biomarkers. However, combining MRI and PET in the same test without considering their inherent structural differences can result in a loss of important information. To address this issue, this paper proposes a novel machine learning framework for combining MRI and PET modalities and a new set of interactions known as Same-Subject-Modalities-Interactions (SSMI) to extract complementary information and new insights. The SSMI relation is derived from MRI and PET and subjected to PCA to construct the SSMI set, which is then concatenated with the other sets. The best set of features is selected and used for classification using Ridge-Classifier. Freesurfer is used to extract measures from 183 ADNI subjects (69 in the AD group and 114 in the CN group), and different classifiers are performed with train-test-split, cross-validation, and validation-set from ADNI-2/GO. The results showed high accuracy, precision, specificity, recall, F1-score, and AUC, with values of 98.94%, 98.27%, 97.10%, 100%, 99.13 and 98.55%, respectively, from ADNI and 98.75%, 98.48%, 93.75%, 100.0%, 99.23%, 96.80% from ADNI2/Go. These results are higher than those achieved by single-modality classification tasks and state-of-the-art approaches. Furthermore, the regions selected by Ridge Classifier are shown to be highly related to Alzheimer's disease biomarkers.

Paper 16

DEMNET: A Deep Learning Model for Early Diagnosis of Alzheimer Diseases and Dementia From MR Images (2020)

Author:Suriya Murugan; Chandran Venkatesan; M. G. Sumithra; Xiao-Zhi Gao; B. Elakkiya;

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is a form of dementia which causes memory, thinking, and behavior disorders in humans. Effective early diagnosis and treatment of AD is of fundamental importance as it can reduce disease progression, allow more effective management of symptoms, facilitate timely patient access to advice and support, and lower associated costs of health care. Given that Alzheimer's typically progresses in stages over an extended period of time, we propose that automated analysis of time sequential data may enhance disease prediction. We present a novel time-series Alzheimer's Disease Prediction Model (ADPM) comprising Random Forest (RF) region of interest (ROI) selection and Gated Recurrent Units (GRU) prediction. Experiments show that our methodology achieves higher classification accuracy in comparison to existing algorithms, and can facilitate prediction of early onset AD. Furthermore, testing demonstrates that random forest ROI selection can identify disease-relative brain regions across different image modalities (MRI, PET, DTI).

Paper 17

Alzheimer's Diseases Detection by Using Deep Learning Algorithms: A Mini-Review (2020)

Author:Suhad Al-Shoukry; Taha H. Rassem; Nasrin M. Makbol

Journal: IEEE Access

Abstract:

The accurate diagnosis of Alzheimer's disease (AD) plays an important role in patient treatment, especially at the disease's early stages, because risk awareness allows the patients to undergo preventive measures even before the occurrence of irreversible brain damage. Although many recent studies have used computers to diagnose AD, most machine detection methods are limited by congenital observations. AD can be diagnosed-but not predicted-at its early stages, as prediction is only applicable before the disease manifests itself. Deep Learning (DL) has become a common technique for the early diagnosis of AD. Here, we briefly review some of the important literature on AD and explore how DL can help researchers diagnose the disease at its early stages.

Paper 18

Anatomical Landmarks and DAG Network Learning for Alzheimer's Disease Diagnosis (2020)

Author: Tian Zhu; Chongfeng Cao; Zhishun Wang; Guangrun Xu; Jianping Qiao

Journal: IEEE Access

Abstract:

The accurate diagnosis and prediction for individuals is crucial in computer-aided diagnosis of Alzheimer's disease (AD). The existing structural magnetic resonance imaging based classification methods of AD diagnosis mainly focus on the voxel level, region level and patch level morphological pattern analysis. However, most of these methods extract features with high dimension which may lead to overfitting problem. Besides, the interaction of different patches is not considered in the classifier ensemble. In this article, we propose a novel anatomical landmarks and directed acyclic graph (DAG) network feature learning based classification algorithm for the diagnosis of AD individuals. First, the anatomical feature patches of gray matter image are identified by the morphological and statistical analysis. Second, a simple and efficient DAG convolutional neural network is proposed to extract the discriminative deep features of image representation. Especially, the deep features are obtained by fusing feature maps of different network levels which contain semantic high-level and high-resolution low-level features. Finally, support vector machine and deep features are utilized to construct the classification model and predict the individual of AD. Experiments on three public datasets including ADNI-1, ADNI-2 and MIRIAD demonstrate that the proposed method can effectively improve the classification performance compared with the state-of-the-art methods for AD diagnosis.

Paper 19

Volumetric Feature-Based Alzheimer's Disease Diagnosis From sMRI Data Using a Convolutional Neural Network and a Deep Neural Network (2021)

Author: Abol Basher; Byeong C. Kim; Kun Ho Lee; Ho Yub Jung

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that is mostly prevalent in people older than 65 years. The hippocampus is a widely studied region of interest (ROI) for a number of reasons, such as memory function analysis, stress development observation and neurological disorder investigation. Moreover, hippocampal volume atrophy is known to be linked with Alzheimer's disease. On the other hand, several biomarkers, such as amyloid beta ($\text{A}\beta_{42}$) protein, tau, phosphorylated tau and hippocampal volume atrophy, are being used to diagnose AD. In this research work, we have proposed a method to diagnose AD based on slice-wise volumetric features extracted from the left and right hippocampi of structural magnetic resonance imaging (sMRI) data. The proposed method is an aggregation of a convolutional neural network (CNN) model with a deep neural network (DNN) model. The left and right hippocampi have been localized automatically using a two-stage ensemble Hough-CNN. The localized hippocampal positions are used to extract ($80 \times 80 \times 80$ voxels) 3-D patches. The 2-D slices are then separated from the 3-D patches along axial, sagittal, and coronal views. The pre-processed 2-D patches are used to extract volumetric features from each slice by using a discrete volume estimation convolutional neural network (DVE-CNN) model. The extracted volumetric features have been used to train and test the classification network. The proposed approach has achieved average weighted classification accuracies of 94.82% and 94.02% based on the extracted volumetric features attributed to the left and right hippocampi, respectively. In addition, it has achieved area under the curve (AUC) values of 92.54% and 90.62% for the left and right hippocampi, respectively. Our method has outperformed the other methods by a certain margin in the same dataset.

Paper 20

Personalized Computer-Aided Diagnosis for Mild Cognitive Impairment in Alzheimer's Disease Based on sMRI and ^{11}C PiB-PET Analysis (2020).

Author: Fatma El-Zahraa Ahmed Abd El-Gamal; Mohammed M. Elmogy; Ashraf Khalil; Mohammed Ghazal; Jawad Yousaf; Xiaolu Qiu; Hassan Soliman; Ahmed Atwan; Hermann B. Frieboes; Gregory N. Barnes; Ayman S. El-Baz.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a neurodegenerative condition that affects the central nervous system and represents 60% to 70% of all dementia cases. Due to an increased aging population, the number of patients diagnosed with AD is expected to exceed 131 million worldwide by 2050. The disease is characterized by various clinical symptoms and pathological features that define three main sequential decline stages, namely, early/mild, intermediate/moderate and late/severe stages. Although it is considered irreversible, early diagnosis of AD is highly desirable to help preserve cognitive function. However, early diagnosis is difficult due to different factors, including the patient-specific development of AD. The main contribution of the proposed work is to present a personalized (i.e., local/brain regional) computer-aided diagnosis (CAD) system for early diagnosis of AD from two perspectives, functional and structural to assist diagnosis. In other words, the proposed system uniquely yields local/regional diagnosis by combining ^{11}C PiB positron emission tomography (^{11}C PiB PET), which provides functional diagnosis, with structural magnetic resonance imaging (sMRI), which provides structural diagnosis. To the best of our knowledge, this is the first work to combine sMRI and the ^{11}C PiB PET for local/regional early diagnosis of AD. The system processes the two modalities through a number of steps: pre-processing, brain labeling (parcellation), feature extraction, and diagnosis. A local/regional diagnosis is presented for each modality separately, followed by the final global diagnosis obtained by integrating the results from the two modalities. Evaluation of the proposed system shows average results of 97.5%, 100%, and 96.77% for accuracy, specificity, and sensitivity, respectively. With further development, it is envisioned that this system could contribute to the early diagnosis of AD in the clinical setting.

Paper: 21

Resting State fMRI and Improved Deep Learning Algorithm for Earlier Detection of Alzheimer's Disease (2020).

Author: Haibing Guo; Yongjin Zhang.

Journal: IEEE Access

Abstract:

The development of computerized healthcare has been powered by diagnostic imaging and machine learning techniques. In particular, recent advances in deep learning have opened a new era in support of multimedia healthcare distribution. For earlier detection of Alzheimer's disease, the study suggested the Improved Deep Learning Algorithm (IDLA) and statistically significant text information. The specific information in clinical

text includes the age, sex and genes of the person and apolipoprotein E; the brain function is established using resting-state functional data (MRI) for the measurement of connectivity in the brain regions. A specialized network of autoencoders is used in earlier diagnosis to distinguish between natural aging and disorder progression. The suggested approach incorporates effectively biased neural network functionality and allows a reliable Alzheimer's disease recognition. In comparison with conventional classifiers depends on time series R-fMRI results, the proposed deep learning algorithm has improved significantly and, in the best cases, the standard deviation reduced by 45%, indicating the forecast model is more reliable and efficient in relation to conventional methodologies. The work examines the benefits of improved deep learning algorithms from recognizing high-dimensional information in healthcare and can lead to the early diagnosis and prevention of Alzheimer's disease.

Paper 22

Relation-Induced Multi-Modal Shared Representation Learning for Alzheimer's Disease Diagnosis(2021)

Author:Zhenyuan Ning; Qing Xiao; Qianjin Feng; Wufan Chen; Yu Zhang

Journal: IEEE Access

Abstract:

The fusion of multi-modal data (e.g., magnetic resonance imaging (MRI) and positron emission tomography (PET)) has been prevalent for accurate identification of Alzheimer's disease (AD) by providing complementary structural and functional information. However, most of the existing methods simply concatenate multi-modal features in the original space and ignore their underlying associations which may provide more discriminative characteristics for AD identification. Meanwhile, how to overcome the overfitting issue caused by high-dimensional multi-modal data remains appealing. To this end, we propose a relation-induced multi-modal shared representation learning method for AD diagnosis. The proposed method integrates representation learning, dimension reduction, and classifier modeling into a unified framework. Specifically, the framework first obtains multi-modal shared representations by learning a bi-directional mapping between original space and shared space. Within this shared space, we utilize several relational regularizers (including feature-feature, feature-label, and sample-sample regularizers) and auxiliary regularizers to encourage learning underlying associations inherent in multi-modal data and alleviate overfitting, respectively. Next, we project the shared representations into the target space for AD

diagnosis. To validate the effectiveness of our proposed approach, we conduct extensive experiments on two independent datasets (i.e., ADNI-1 and ADNI-2), and the experimental results demonstrate that our proposed method outperforms several state-of-the-art methods

DOI: 10.1109/TMI.2021.3063150.

Paper 23:

Reservoir Computing for Early Stage Alzheimer's Disease Detection (2022).

Author: Nickson Mwamsojo; Frederic Lehmann; Mounim A. El Yacoubi; Kamel Merghem et al.

Journal: IEEE Access

Abstract:

Artificial Neural Networks (ANNs) have amassed unprecedented success in information processing ranging from image recognition to time series prediction. The success can largely be attributed to the availability of large datasets for training and the increased complexity of the models. Unfortunately, for some applications only a limited amount of samples is available for training. Fewer training samples increases the risk of over-fitting and poor generalization especially in high complexity models. Moreover, complex models with a large number of trainable parameters require more energy to train and optimize compared to simpler ones. In this paper, to the best of our knowledge, we propose the first use of ANNs for Early Stage Alzheimer Disease classification (ES-AD) from the handwriting (HW). We propose using a framework for building Recurrent Neural Networks (RNNs) known as Reservoir Computing (RC), both numerically and experimentally, that simplifies training by optimizing the output layer only. We also propose the Bidirectional Long Term Short Term (BiLSTM) and Convolutional Neural Network (CNN) methods for comparison. For a fairer comparison, we not only consider the accuracies but also the energy costs incurred to obtain the respective accuracies in order to assess the accuracy-efficiency trade-off. Our numerical and experimental results show that RC yields a classification accuracy of 85%, which is 3% worse than that of BiLSTM and 2% better than that of CNN, at a relatively lower training and significantly lower inference costs. We hope that our findings highlight the importance of examining the accuracy-efficiency trade-off of various models in the community in order to reduce the overall impact of ANNs training on the environment.

Paper 24:

An Improved LeNet-Deep Neural Network Model for Alzheimer's Disease Classification Using Brain Magnetic Resonance Images (2021)

Author: Ruhul Amin Hazarika; Ajith Abraham; Debdatta Kandar; Arnab Kumar Maji

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is a psychological disorder in elderly people which causes severe intellectual disabilities. Proper processing of neuro-images can provide differences in brain tissues which may help in diagnosing the disease more effectively. But, due to the complex structures, this is a challenge in differentiating the brain tissues and classifying AD using traditional classification mechanisms. Deep Neural Network (DNN) is a machine learning technique that has the ability to absorb the most important information for classifying an object accurately. LeNet is a popular DNN based model with a simple and effective architecture that also consumes very less implementation time. As like most of the DNN models, LeNet also uses MaxPooling layer for dimensionality reduction by eliminating the information of minimum valued elements. In brain images low intensity valued pixels also may contain very important features. To keep the minimum valued elements too in the network, we have created a separate layer that performs Min-Pooling operation. MinPooling and MaxPooling layers are then concatenated together. Finally, we have replaced all MaxPooling Layers in LeNet by the concatenated layers. We have analysed and compared the performances of modified LeNet model with 20 other most commonly used DNN models, and some of the related works. It is observed that, the modified LeNet model achieved the highest performances. It is also observed that, original LeNet model can classify AD with a performance rate of 80%, whereas, the proposed modified LeNet model achieved an average performance rate of 96.64%.

Paper 25

Conv-eRVFL: Convolutional Neural Network Based Ensemble RVFL Classifier for Alzheimer's Disease Diagnosis(2023)

Author:Rahul Sharma; Tripti Goel; M. Tanveer; P. N. Suganthan; Imran Razzak; R. Murugan.

Journal: IEEE Access

Abstract:

As per the latest statistics, Alzheimer's disease (AD) has become a global burden over the following decades. Identifying AD at the intermediate stage became challenging, with mild cognitive impairment (MCI) utilizing credible biomarkers and robust learning approaches. Neuroimaging techniques like magnetic resonance imaging (MRI) and positron emission tomography (PET) are practical research approaches that provide structural atrophies and metabolic variations. With the help of MRI and PET scans, metabolic and structural changes in AD patients can be visible even ten years before the disease's onset. This paper proposes a novel wavelet packet transform-based structural and metabolic image fusion approach using MRI and PET scans. An eight-layer trained CNN extracts features from multiple layers and these features are fed to an ensemble of non-iterative random vector functional link (RVFL) models. The RVFL network incorporates the μ -membership fuzzy function as an activation function that helps overcome outliers. Lastly, outputs of all the customized RVFL classifiers are averaged and fed to the RVFL classifier to make the final decision. Experiments are performed over Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, and classification is made over CN vs. AD vs. MCI. The model performance obtained is decent enough to prove the effectiveness of the fusion-based ensemble approach.

Dharshini E

LITERATURE REVIEW

Paper 26:

Multimodal 3D Deep Learning for Early Diagnosis of Alzheimer's Disease (2024)

Authors:Seung Kyu Kim; Quan Anh Duong; Jin Kyu Gahm.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a neurodegenerative disease that affects the elderly and leads to cognitive decline and memory loss. Treatments for stopping or slowing the progression of AD have not been discovered yet; therefore, delaying the progression of AD is the only option, which makes early diagnosis of AD crucial. Additionally, although A β plaques and tau proteins are considered the causes of early AD, few studies have used this information to diagnose early AD. In this study, a middle-fusion multimodal model is proposed for the diagnosis of early AD. The proposed multimodal model extracts features without loss using a depthwise separable convolution block without an activation function. Subsequently, middle fusion is applied using mix skip connection and sharing weight convolution blocks, both designed to learn the complex relationships between modalities. In contrast to other studies, the proposed approach has three main novelties. 1) A middle-fusion multimodal model is proposed for the early diagnosis of AD. 2) The proposed model is evaluated using the entire ADNI series, including T1-weighted magnetic resonance imaging (T1w MRI) and 18F-FluoroDeoxyGlucose positron emission tomography (FDG PET) from the ADNI1 dataset, as well as A β PET and tau protein PET from ADNI2 and ADNI3 datasets. 3) A novel region-of-interest (ROI) extraction method is proposed for the hippocampus, middle temporal, and inferior temporal regions, which are known to be affected in the early stages of AD. In the experimental results, the proposed multimodal model achieved a balanced accuracy of 1.00, for the task of Alzheimer's disease vs cognitive normal (CN) and 0.76 for the task of mild cognitive impairment vs cognitive normal.

Paper 27:

Longitudinal Alzheimer's Disease Progression Prediction With Modality Uncertainty and Optimization of Information Flow (2025)

Authors :Duy-Phuong Dao; Hyung-Jeong Yang; Jahae Kim; Ngoc-Huynh Ho

Abstract:

Alzheimer's disease (AD) is a global neurodegenerative disorder that affects millions of individuals worldwide. Actual AD imaging datasets challenge the construction of reliable longitudinal models owing to imaging modality uncertainty. In addition, they are still unable to retain or obtain important information during disease progression from previous to followup time points. For example, the output values of current gates in recurrent models should be close to a specific value that indicates the model is uncertain about retaining or forgetting information. In this study, we propose a model which can extract and constrain each modality into a common representation space to capture intermodality interactions among different modalities associated with modality uncertainty to predict AD progression. In addition, we provide an auxiliary function to enhance the ability of recurrent gate robustly and effectively in controlling the flow of information over time using longitudinal data. We conducted comparative analysis on data from the Alzheimer's Disease Neuroimaging Initiative database. Our model outperformed other methods across all evaluation metrics. Therefore, the proposed model provides a promising solution for addressing modality uncertainty challenges in multimodal longitudinal AD progression prediction

Paper 28:

Optimized One vs One Approach in Multiclass Classification for Early Alzheimer's Disease and Mild Cognitive Impairment Diagnosis (2020)

Author: Carmen Jiménez-Mesa, Ignacio A. Illan, Alberto Martín-Martín, Diego Castillo-Barnes

Journal: IEEE Access

Abstract:

The detection of Alzheimer's Disease in its early stages is crucial for patient care and drugs development. Motivated by this fact, the neuroimaging community has extensively applied machine learning techniques to the early diagnosis problem with promising results. The organization of challenges has helped the community to address different raised problems and to standardize the approaches to the problem. In this work we use the data from international challenge for automated prediction of MCI from MRI data to address the multiclass classification problem. We propose a novel multiclass classification approach that addresses the outlier detection problem, uses pairwise t-test feature selection, project the selected features onto a Partial-Least-Squares multiclass subspace, and applies one-versus-one error correction output codes classification. The proposed method yields to an accuracy of 67% in the multiclass classification, outperforming all the proposals of the competition.

Paper 29:

An Ensemble-Learning Based Application to Predict the Earlier Stages of Alzheimer's Disease (AD) (2020)

Author:Asif Hassan Syed, Tabrej Khan, Atif Hassan, Nashwan A. Alromema, Muhammad Binsawad, and Alhuseen Omar Alsayed

Journal: IEEE Access

Abstract:

The fact that ensemble methods enhance the prediction performance. Therefore, we focused on developing a weighted ensemble method using a novel combination of Cerebrospinal Fluid (CSF) protein biomarkers to predict AD's earlier stages with greater accuracy than the state-of-the-art CSF protein biomarkers. In this regard, two feature selection methods, namely the Recursive Feature Elimination (RFE) and L1 regularization method were used to screen the most important subset of features for building a classification model using the Mild Cognitive Impairment (MCI) dataset. A novel combination of three biomarkers, namely Cystatin C, Matrix metalloproteinases (MMP10), and tau protein, was screened using the linear Support Vector Machine (SVM) and Logistic Regression (LR) classifier based RFE method. Two-tailed unpaired t-test analysis at a 5% significance level showed a significant difference between the mean levels of Cystatin C, MMP10, and tau protein between cognitive normal and cognitively impaired groups. An ensemble model using a weighted average of two best performing classifiers (LR and Linear SVM) was created using a novel subset of three

most informative features. Our ensemble model's weighted average results performed significantly better than LR and Linear SVM base classifiers' performance. The Receiver Operating Characteristic Curve (ROC_AUC) and Area under Precision-Recall values (AUPR) of our proposed model were observed to be 0.9799 ± 0.055 0.9108 ± 0.015 , respectively. The performance of our proposed weighted averaged ensemble model built using a novel combination of CSF protein biomarkers was significantly better ($p < 0.001$) than models generated using different combinations of CSF protein biomarkers obtained from recent studies. An ensemble-learning based application was implemented and deployed at Heroku at <https://appsalzheimer.herokuapp.com>.

Paper 30:

A Novel Convolutional Variation of Broad Learning System for Alzheimer's Disease Diagnosis by Using MRI Images (2020)

Author: Ruizhi Han; C. L. Philip Chen; Zhulin Liu

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a serious chronic health problem that causes great pain and loss to patients and their families. Its early and accurate diagnosis would achieve significant progress on the prevention and treatment of the disease. Magnetic Resonance Imaging (MRI) is a commonly used technique in nuclear medical diagnostics. However, it is still a challenging problem to diagnose AD, Control Normal (CN), and Mild Cognitive Impairment (MCI) because of the complex structures of MRI. In this paper, diagnosing models for MRI images are proposed to identify the various stages of AD based on the Broad Learning Systems (BLS), as well as its convolutional variants. To verify the validity of the proposed models, experiments on MRI images collected from the ADNI website are tested and evaluated. The results show that our algorithms outperform the other state-of-the-art algorithms for various tasks with better accuracy and less training times. Finally, the cross-domain learning ability of the proposed algorithms is verified on an independent AD dataset.

Paper 31:

Longitudinal Brain MR Image Modeling Using Personalized Memory for Alzheimer's Disease (2021).

Author: Seong Tae Kim; Umut Kucukaslan; Nassir Navab

Journal: IEEE Access

Abstract:

Longitudinal analysis of a disease is an important issue to understand its progression and design prognosis and early diagnostic tools. From the longitudinal images where data is collected from multiple time points, both the spatial structural information and the longitudinal variations are captured. The temporal dynamics are more informative than static observations of the symptoms, particularly for neurodegenerative diseases such as Alzheimer's disease, whose progression spans over the years with early subtle changes. In this paper, we propose a new generative framework to predict the lesion progression over time. Our method first encodes images into the structural and longitudinal state vectors, where interpolation or extrapolation of feature vectors in the time axis can be performed for the manipulation of these feature vectors. These processed feature vectors can be decoded into image space to predict the image at the time point which we are interested in. During the training, we force the model to encode longitudinal changes into longitudinal state features and capture the structural information in a separate vector. Moreover, we introduce a personalized memory for the online update scheme, which adapts the model to the target subject, which helps the model preserve fine details of brain image structures in each subject. Experimental results on the public longitudinal brain magnetic resonance imaging dataset show the effectiveness of the proposed method.

Paper 32:

Computational Modeling of Dementia Prediction Using Deep Neural Network: Analysis on OASIS Dataset (2021)

Author: . Shakila Basheer, Surbhi Bhatia, and Sapiah Binti Sakri

Journal: IEEE Access

Abstract:

Alzheimer is a progressive disease and it is the most prevalent neurodegenerative disorder. It is believed that the people with mild cognitive impairment are at high risk of developing this disease. According to the annual report released by the Alzheimer's Association®2020, Alzheimer is the sixth leading cause of death in the United States. Thus, there is a need of educating people about this disease, reducing the risks by militating the necessary precautions to disseminate its affect by diagnosing it at early stages. It is also important to propose some recent advancement in this research which can help in early prediction of the disease using machine learning techniques. This paper intends to develop the novel algorithm by proposing changes in the designing of capsule network for best prediction results and making the model computationally efficient. The research is conducted on the Open Access Series of Imaging Studies (OASIS) dataset with dimensions (373 X 15) to diagnose the labels into two groups, as demented and non-demented. The novelty lies in conducting the in-depth research in identifying the importance of features, correlation study between factors and density of data showing status of factors by studying hierarchical examination of all the data points available using exploratory data analysis. Several optimization functions are conducted on the variables and feature selection is done to make the model faster and more accurate. The claims have been validated by showing the correlation accuracy at several iterations and layers with an admissible accuracy of 92.39%. The model is compared with state-of-art deep learning classifiers taken as benchmarks using different performance metrics. The ablation study is conducted on the proposed model using OASIS dataset to justify the predictions of the model

Paper 33:

AlzheimerNet: An Effective Deep Learning Based Proposition for Alzheimer's Disease Stages Classification From Functional Brain Changes in Magnetic Resonance Images (2023)

Author:Yonglin Chen; Huabin Wang; Gong Zhang; Xiao Liu; Wei Huang; Xianjun Han; Xuejun Li; Melanie Martin; Liang Tao.

Journal: IEEE Access

Abstract:

Alzheimer's disease is largely the underlying cause of dementia due to its progressive neurodegenerative nature among the elderly. The disease can be divided into five stages: Subjective Memory Concern (SMC), Mild Cognitive Impairment (MCI), Early MCI (EMCI), Late MCI (LMCI), and Alzheimer's Disease (AD). Alzheimer's disease is conventionally diagnosed using an MRI scan of the brain. In this research, we propose a fine-tuned convolutional neural network (CNN) classifier called AlzheimerNet, which can identify all five stages of Alzheimer's disease and the Normal Control (NC) class. The ADNI database's MRI scan dataset is obtained for use in training and testing the proposed model. To prepare the raw data for analysis, we applied the CLAHE image enhancement method. Data augmentation was used to remedy the unbalanced nature of the dataset and the resultant dataset consisted of 60000 image data on the 6 classes. Initially, five existing models including VGG16, MobileNetV2, AlexNet, ResNet50 and InceptionV3 were trained and tested to achieve test accuracies of 78.84%, 86.85%, 78.87%, 80.98% and 96.31% respectively. Since InceptionV3 provides the highest accuracy, this model is later modified to design the AlzheimerNet using RMSprop optimizer and learning rate 0.00001 to achieve the highest test accuracy of 98.67%. The five pre-trained models and the proposed fine-tuned model were compared in terms of various performance matrices to demonstrate whether the AlzheimerNet model is in fact performing better in classifying and detecting the six classes. An ablation study shows the hyperparameters used in the experiment. The suggested model outperforms the traditional methods for classifying Alzheimer's disease stages from brain MRI, as measured by a two-tailed Wilcoxon signed-rank test, with a significance of <0.05 .

Paper 34:

Automated Detection of Alzheimer's Disease and Mild Cognitive Impairment Using Whole Brain MRI (2022)

Author: Fazal Ur Rehman Faisal and Goo-Rak Kwon

Journal: IEEE Access

Abstract

Early diagnosis is critical for the development and success of interventions, and neuroimaging is one of the most promising areas for early detection of Alzheimer's disease (AD). This study is aimed to develop a deep learning method to extract valuable AD biomarkers from structural magnetic resonance imaging (sMRI) and classify brain images into AD, mild cognitive impairment (MCI), and cognitively normal (CN) groups. In this work, we adapted and trained convolutional neural networks (CNNs) on sMRI images of the brain from ADNI datasets available in online databases. Our proposed mechanism was used to combine features from different layers to hierarchically transform the images from magnetic resonance imaging into more compact high-level features. The proposed method has a reduced number of parameters which reduces the computation complexity. The method is compared with the existing state-of-the-art works for AD classification, which show superior results for the widely used evaluation metrics, including accuracy, an area under the ROC curve, etc., suggesting that our proposed convolution operation is suitable for the AD diagnosis

Paper 35:

sMRI-PatchNet: A Novel Efficient Explainable Patch-Based Deep Learning Network for Alzheimer's Disease Diagnosis With Structural MRI (2024)

Author: : Xin Zhang; Liangxiu Han; Lianghao Han; Haoming Chen; Darren Dancey; Daoqiang Zhang.

Journal: IEEE Access

Abstract:

Structural magnetic resonance imaging (sMRI) can identify subtle brain changes due to its high contrast for soft tissues and high spatial resolution. It has been widely used in diagnosing neurological brain diseases, such as Alzheimer's disease (AD). However, the size of 3D high-resolution data poses a significant challenge for data analysis and processing. Since only a few areas of the brain show structural changes highly associated with AD, the patch-based methods dividing the whole data into several regular patches have shown promising for more efficient image analysis. The major challenges of the patch-based methods include identifying the discriminative patches,

combining features from the discrete discriminative patches, and designing appropriate classifiers. This work proposes a novel efficient patch-based deep learning network (sMRI-PatchNet) with explainable patch localisation and selection for AD diagnosis. Specifically, it consists of two primary components: 1) A fast and efficient explainable patch selection method for determining the most discriminative patches; and 2) A novel patch-based network for extracting deep features and AD classification with position embeddings to retain position information, capable of capturing the global and local information of inter- and intra-patches. This method has been applied for the AD classification and the prediction of the transitional state moderate cognitive impairment (MCI) conversion with real datasets. The experimental evaluation shows that the proposed method can identify discriminative pathological locations effectively with a significant reduction on patch numbers used, providing better performance in terms of accuracy, computing performance, and generalizability, in contrast to the state-of-the-art methods

Paper 36:

An Exemplar Pyramid Feature Extraction Based Alzheimer Disease Classification Method (2022)

Author: Heba Soliman Zaina; Samir Brahim Belhaouari; Tanya Stanko; Vladimir Gorovoy

Journal: IEEE Access

Abstract:

Dementia is a term used to describe a variety of symptoms related to cognitive impairment in which Alzheimer disease represents 60% – 70% of the cases. As of today, there is no cure for this disease and the only way to prevent any associated medical, economic, and financial impacts or losses is to detect the disease early and work closely with suspected patients to prevent any further progress. In this research, a methodology consisting of 4 modules is proposed: (1) preprocessing, exemplar pyramid along with bi-linear interpolation followed by (2) feature extraction using Gray Level Co-Occurrence Matrix and Local Binary Pattern then (3) concatenation of all extracted features and finally (4) classification of Alzheimer disease stage using deep learning, Multi-Layer Perceptron, in particular. Our proposed method was tested using the

MPRAGE structural MRI dataset from Alzheimer Disease Neuro Imaging Initiative (ADNI), and it outperformed other techniques used in the literature review. An accuracy result of 89.80 was reported for multi-class classification of 4 stages of Alzheimer disease (Cognitive Normal, Early Mild Cognitive Impairment, Late Mild Cognitive Impairment and Alzheimer Disease) for both Gray Matter (GM) and White Matter (WM). In term of binary-class classification, we were able to achieve very good results using both GM and WM. By using GM, we were able to distinguish between CN vs EMCI, EMCI vs AD and LMCI vs AD with accuracy results of 96.43%, 90.91% and 95.24% respectively. And using WM, we were able to distinguish between CN vs LMCI with 100% accuracy and EMCI vs LMCI with 95.65% accuracy. While we achieved the same accuracy result of 96.15 using both WM and GM.

Paper 37:

Automatic Early Diagnosis of Alzheimer's Disease Using 3D Deep Ensemble Approach (2022)

Author: Aya Gamal, Mustafa Elattar, and Sahar Selim

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is considered the 6th leading cause of death worldwide. Early diagnosis of AD is not an easy task, and no preventive cures have been discovered yet. Having an accurate computer-aided system for the early detection of AD is important to help patients with AD. This study proposes a new approach for classifying disease stages. First, we worked on the MRI images and split them into an appropriate format to avoid data leakage. Subsequently, a simple and fast registration-free preprocessing pipeline was applied to the dataset. Numerous experiments were conducted to analyze the performances of different 3D classification architectures. Finally, an ensemble learning approach is applied to the top-performing models. The outstanding performance of the proposed method was demonstrated using augmentation of the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset. Our proposed ensemble approach outperforms studies in literature for distinguishing between people with AD and mild cognitive impairment (MCI), and MCI and cognitive normal (CN) with an AUC

score of 91.28% and 88.42%, respectively. We also targeted the multiclass task, which was marginalized in previous work, by differentiating between the three stages of the disease.

Paper 38:

HSI-LFS-BERT: Novel Hybrid Swarm Intelligence Based Linguistics Feature Selection and Computational Intelligent Model for Alzheimer's Prediction Using Audio Transcript (2022)

Author:Yusera Farooq Khan, Baijnath Kaushik, Mohammad Khalid Imam Rahmani, and Md. Ezaz Ahmed

Journal: IEEE Access

Abstract:

Alzheimer's dementia (AD) affects memory, language, and cognition and worsens over time. Therefore, it is critical to develop a reliable method for early detection of permanent brain atrophy and cognitive impairment. This study used clinical transcripts, a text-based adaptation of the original audio recordings of Alzheimer's patients. This audio transcript data were taken from DementiaBank, which is the largest public dataset of AD transcripts. This study aims to show how Transfer Learning-based models and swarm intelligence optimization techniques can be used to predict Alzheimer's disease. To enhance the prediction performance for Alzheimer's disease, a hybrid swarm intelligence linguistic feature selection (HSI-LFS) approach is proposed that extracts a combined feature set using Particle Swarm Optimization (PSO), Dragonfly Optimization (DO), and Grey Wolf Optimization (GWO) algorithms. In addition, a transfer learning-based model called HSI-LFS-BERT, a combination of the HSI-LFS feature selection method and Bidirectional Encoder Representations from Transformer (BERT) algorithm, is proposed. The proposed model was compared using two feature sets: the first set consisted of the initial feature set and the second set contained a hybrid feature set that was extracted using the suggested HSI-LFS method. BERT embedding with

HSI-LFS outperformed the conventional feature set, providing the most accurate modeling parameters while reducing the computations by 27.19%. The proposed HSI-LFS-BERT model outperformed state-of-the-art models, achieving 98.24% accuracy, 91.56% precision, and 98.78% recall.

Paper 39:

Deviation from Model of Normal Aging in Alzheimer's Disease: Application of Deep Learning to Structural MRI Data and Cognitive Tests (2022)

Author: Tetiana Habuza; Nazar Zaki; Elfadil A. Mohamed; Yauhen Statsenko

Journal: IEEE Access

Abstract:

Background. Psychophysiological and cognitive tests as well as other functional studies can detect pre-symptomatic stages of dementia. When assembled with structural data, cognitive tests diagnose NDs more reliably thus becoming a multimodal diagnostic tool. Objective. Our main goal is to improve screening for dementia by studying an association between the brain structure and its function. Hypothetically, the brain structure-function association has features specific for either disease-related cognitive deterioration or normal neurocognitive slowing while aging. Materials and methods. We studied a total number of 287 cognitively normal cases, 646 of mild cognitive impairment, and 369 of Alzheimer's disease. To work out a new marker of neurodegeneration, we created a convolutional neural network-based regression model and predicted the cognitive status of the cognitively preserved examinee from the brain MRI data. This was a model of normal aging. A big deviation from the model suggests a high risk of accelerated cognitive decline. Results. The deviation from the model of normal aging can accurately distinguish cognitively normal subjects from MCI patients (AUC = 0.9957). We also achieved creditable performance in the MCI-versus-AD classification (AUC = 0.9793). We identified a considerable difference in the MMSE test between A-positive and A-negative demented individuals according to ATN-criteria (6.27 ± 1.82 vs 5.32 ± 1.9 ; $p < 0.05$). Conclusion. The deviation from the model of normal aging can be potentially used as a marker of dementia and as a tool for differentiating

Alzheimer's disease from non-Alzheimer's dementia. To find and justify a reliable threshold levels, further research is required .

Paper 40:

Diagnosis of Alzheimer's Disease Using Convolutional Neural Network With Select Slices by Landmark on Hippocampus in MRI Images (2023).

Author: Yori Pusparani; Chih-Yang Lin; Yih-Kuen Jan; Fu-Yu Lin; Ben-Yi Liao; Peter Ardianto; Isack Farady; John Sahaya Rani Alex; Jeetashree Aparajeeta; Wen-Hung Chao;

Journal: IEEE Access

Alzheimer's disease (AD) is a major public health priority. Hippocampus is one of the most affected areas of the brain and is easily accessible as a biomarker using MRI images in machine learning for diagnosing AD. In machine learning, using entire MRI image slices showed lower accuracy for AD classification. We present the select slices method by landmarks on the hippocampus region in MRI images. This study aims to see which views of MRI images have higher accuracy for AD classification. Then, to get the value of three views and categories, we used multiclass classification with the publicly available Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset using Resnet50 and LeNet. The models were used in a total dataset of 4,500 MRI slices in three views and categories. Our study demonstrated that the selecting slices performed better than using entire slices in MRI images for AD classification. Our method improves the accuracy of machine learning, and the coronal view showed higher accuracy. This method played a significant role in improving the accuracy of machine learning performance. The results for the coronal view were similar to the medical experts usually used to diagnose AD. We also found that LeNet models became the potential model for AD classification.

Paper 41:

DICE-Net: A Novel Convolution-Transformer Architecture for Alzheimer Detection in EEG Signals (2023)

Author: Andreas Miltiadous, Emmanouil Gionanidis, Katerina D. Tzimourta, Nikolaos Giannakeas, and Alexandros T. Tzallas

Journal: IEEE Access

Abstract:

jective: Alzheimer's disease (AD) is a progressive neurodegenerative disorder that affects a significant percentage of the elderly. EEG has emerged as a promising tool for the timely diagnosis and classification of AD or other dementia types. This paper proposes a novel approach to AD EEG classification using a Dual-Input Convolution Encoder Network (DICE-net). Approach: Recordings of 36 AD, 23 Frontotemporal dementia (FTD), and 29 age-matched healthy individuals (CN) were used. After denoising, Band power and Coherence features were extracted and fed to DICE-net, which consists of Convolution, Transformer Encoder, and Feed-Forward layers. Main results: Our results show that DICE-net achieved an accuracy of 83.28% in the AD-CN problem using Leave-One-Subject-Out validation, outperforming several baseline models, and achieving good generalization performance. Significance: Our findings suggest that a convolution transformer network can effectively capture the complex features of EEG signals for the classification of AD patients versus control subjects and may be expanded to other types of dementia, such as FTD. This approach could improve the accuracy of early diagnosis and lead to the development of more effective interventions for AD

Paper 42:

Classification of Alzheimer's Disease Using Ensemble Convolutional Neural Network With LFA Algorithm (2024)

Author: Chang-Min Kim; Woobeom Lee

Journal: IEEE Access

Abstact:

Alzheimer's disease (AD) is a disease that develops gradually, ultimately causing deterioration of brain functions. Thus, early diagnosis is essential for treating and managing AD. Magnetic-resonance-imaging (MRI)-based AD diagnosis classifies the stage of AD according to the extent of atrophy caused to a patient's hippocampal and entorhinal cortex. In this case, the shape of the patient's brain serves as a crucial feature. Therefore, in this paper, we propose an ensemble convolutional neural network (CNN) model that can classify the AD stage according to the shape of a patient's brain. The proposed model is structured by combining a convolutional layer part of the visual geometry group network (VGGNet) model, with proven performance in image classification, and a 1D CNN model into a pipeline. Here, the 1D CNN applies the line segment feature analysis (LFA) algorithm to MRI images to transform the visual line segment information of the images into vectors and record strong features indicating the shape of the brain. This is followed by 1D CNN model training. Notably, the 1D CNN model can carefully observe the brain shape owing to the parallel connection of ten 1D convolutional layers with LFA features. Subsequently, the brain shape information is combined with features obtained from the original image through the VGGNet to improve the model performance compared to that of existing methods. To evaluate the performance of the proposed ensemble CNN model, MRI datasets collected from Kaggle are used to evaluate and compare the proposed model with existing image classification methods and methods proposed in related studies. The experimental results reveal that the proposed model demonstrates superior performance with an accuracy of 0.986 and a loss of 0.0385.

Paper 43:

Robust Hippocampus Localization From Structured Magnetic Resonance Imaging Using Similarity Metric Learning (2022)

Author: Samsuddin Ahmed; Kun Ho Lee; Ho Yub Jung

Journal: IEEE Access

Accurate demarcation of anatomical landmarks in 3D medical imaging is a safety-critical and challenging task. State-of-the-art approaches formulate landmark localization either as a classification or as a regression problem. In this study, feature classification is

performed as a verification step in a cascaded Hough regression networks (HRNs) for hippocampus localization in the structured magnetic resonance images of the brain. Global and local features of the landmarks are learned with coarse prediction and fine-tuning convolutional neural networks for coarse-to-fine localization. Siamese network was trained to learn a deep metric for verifying the roughly estimated locations. Feature verification with the Siamese network drops the outlier predictions and increase the robustness in prediction. Three-view patches(TVPs) with a size of $64 \times 64 \times 3$ are fed for rough estimation while the TVP sizes for Siamese-based verification and Hough regression network (HRN)-based fine-grained estimations are $32 \times 32 \times 3$ and $16 \times 16 \times 3$, respectively. The experiment was performed on the Gwangju Alzheimer's and Related Dementia's (GARD) cohort data set. The proposed approach demonstrated better performance with the errors of 1.70 ± 0.50 millimeters(mm) and 1.66 ± 0.49 mm for localizing the left and right hippocampi in the GARD data set. In Alzheimer's Disease Neuroimaging Initiative (ADNI) data set, the observed errors were 1.79 ± 0.83 mm and 1.55 ± 0.61 mm for localizing left and right hippocampus, respectively. Our results are comparable to those obtained by the state-of-the-art methods. .

Paper 44:

An Empirical Analysis of Diffusion, Autoencoders, and Adversarial Deep Learning Models for Predicting Dementia Using High-Fidelity MRI (2024)

Author: Pranshav Gajjar; Manav Vikas Garg; Shivani Desai; Hitesh Chhinkaniwala; Harshal A. Sanghvi; Riki H. Patel; Shailesh Gupta; Abhijit S. Pandya

Journal: IEEE Access

Abstract:

This study explores cutting-edge computational technologies and intelligent methods to create realistic synthetic data, focusing on dementia-centric Magnetic Resonance Imaging (MRI) scans related to Alzheimer's and Parkinson's diseases. The research delves into Generative Adversarial Networks (GANs), Variational Autoencoders, and Diffusion Models, comparing their efficacy in generating synthetic MRI scans. Using datasets from Alzheimer's and Parkinson's patients, the study reveals intriguing findings. In the Alzheimer dataset, diffusion models produced non-dementia images with the lowest Frechet Inception Distance (FID) score at 92.46, while data-efficient GANs

excelled in generating dementia images with an FID score of 178.53. In the Parkinson dataset, data-efficient GANs achieved remarkable FID scores of 102.71 for dementia images and 129.77 for non-dementia images. The study also introduces a novel aspect by incorporating a classification study, validating the generative metrics. DenseNets, a deep learning architecture, exhibited superior performance in disease detection compared to ResNets. Training both models on images generated by diffusion models further improved results, with DenseNet achieving accuracies of 80.84% and 92.42% in Alzheimer's and Parkinson's disease detection, respectively. The research not only presents innovative generative architectures but also emphasizes the importance of classification metrics, providing valuable insights into the synthesis and detection of neurodegenerative diseases through advanced computational techniques.

Paper 45:

A Cross-Modal Mutual Knowledge Distillation Framework for Alzheimer's Disease Diagnosis: Addressing Incomplete Modalities (2025).

Author:Min Gu Kwak; Lingchao Mao; Zhiyang Zheng; Yi Su; Fleming Lure; Jing Li.

Journal: IEEE Access

Abstract:

Early detection of Alzheimer's Disease (AD) is crucial for timely interventions and optimizing treatment outcomes. Integrating multimodal neuroimaging datasets can enhance the early detection of AD. However, models must address the challenge of incomplete modalities, a common issue in real-world scenarios, as not all patients have access to all modalities due to practical constraints such as cost and availability. We propose a deep learning framework employing Incomplete Cross-modal Mutual Knowledge Distillation (IC-MKD) to model different sub-cohorts of patients based on their available modalities. In IC-MKD, the multimodal model (e.g., MRI and PET) serves as a teacher, while the single-modality model (e.g., MRI only) is the student. Our IC-MKD framework features three components: a Modality-Disentangling Teacher (MDT) model designed through information disentanglement, a student model that

learns from classification errors and MDT's knowledge, and the teacher model enhanced via distilling the student's single-modal feature extraction capabilities. Moreover, we show the effectiveness of the proposed method through theoretical analysis and validate its performance with simulation studies. In addition, our method is demonstrated through a case study with Alzheimer's Disease Neuroimaging Initiative (ADNI) datasets, underscoring the potential of artificial intelligence in addressing incomplete multimodal neuroimaging datasets and advancing early AD detection. **Note to Practitioners**—This paper was motivated by the challenge of early AD diagnosis, particularly in scenarios when clinicians encounter varied availability of patient imaging data, such as MRI and PET scans, often constrained by cost or accessibility issues. We propose an incomplete multimodal learning framework that produces tailored models for patients with only MRI and patients with both MRI and PET. This approach improves the accuracy and effectiveness of early AD diagnosis, especially when imaging resources are limited, via bi-directional knowledge transfer. We introduced a teacher model that prioritizes extracting common information between different modalities, significantly enhancing the student model's learning process. This paper includes theoretical analysis, simulation study, and real-world case study to illustrate the method's promising potential in early AD detection. However, practitioners should be mindful of the complexities involved in model tuning. Future work will focus on improving model interpretability and expanding its application. This includes developing methods to discover the key brain regions for predictions, enhancing clinical trust, and extending the framework to incorporate a broader range of imaging modalities, demographic information, and clinical data. These advancements aim to provide a more comprehensive view of patient health and improve diagnostic accuracy across various neurodegenerative diseases

Paper 46:

Design of a CNN–Swin Transformer Model for Alzheimer's Disease Prediction Using MRI Images (2025)

Author:Velu K. and N. Jaisankar

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is a progressive neurological condition that deteriorates memory, cognition, and behavior, especially in older adults. Timely identification is essential to enhance patient outcomes and inform therapy choices. This research introduces an extensive deep learning model for multiclass Alzheimer's disease stage classification using structural MRI images from the publicly accessible OASIS dataset. The procedure begins with thorough preprocessing, including skull stripping, axial slice extraction, and intensity normalization to guarantee uniform input quality. Deep Convolution Generative Adversarial Network (DCGAN) is used to produce authentic synthetic MRI slices, therefore addressing data imbalance and enhancing class representation and training stability. The EffSwin-XNet model introduces a novel hybrid deep learning framework that strategically fuses EfficientNet-B0 and the Swin Transformer, enabling both local and global feature extraction from MRI brain images. This represents a significant advancement over conventional convolutional neural networks and a feature fusion attention mechanism that adaptively emphasizes discriminative features. Grad-CAM is used for explainability to view the brain areas influencing each categorization judgment, hence increasing therapeutic confidence. The model attains a classification accuracy of 95.3%, surpassing traditional CNN and hybrid benchmarks. This study presents an enhanced, interpretable, and efficient method for stage-wise categorization of Alzheimer's Disease, demonstrating significant potential for use in clinical decision-support systems for early classification and intervention.

Paper 47:

Multimodal Fusion-Based Deep Learning Network for Effective Diagnosis of Alzheimer's Disease (2022)

Author:Shubham Dwivedi; Tripti Goel; M. Tanveer; Murugan Raman; Rahul Sharma.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a prevalent, irreversible, chronic, and degenerative disorder whose diagnosis at the prodromal stage is critical. Mostly, single modality data, such as

magnetic resonance imaging (MRI) or positron emission tomography (PET), are used to make predictions in AD studies. However, the metabolic and structural data fusion can provide a holistic view of AD-staging analysis. To achieve this objective, a novel multimodal fusion-based method is proposed in this article. An optimal fusion of MRI and PET is achieved by harnessing demon algorithm and discrete wavelet transform. Finally, the fused image features are extracted using ResNet-50, and these features are classified using robust energy least square twin support vector machine classifier. Experiments on the AD neuroimaging initiative dataset show descent accuracy of 97%, 94%, and 97.5% for cognitive normal (CN) versus AD, CN versus mild cognitive impairment (MCI), and AD versus MCI, respectively. The proposed model will be beneficial for health professionals in accurately diagnosing AD at an early stage

Paper 48:

Automated Detection of Alzheimer's Disease: A Multi-Modal Approach With 3D MRI and Amyloid PET (2024)

Author:. Giovanna Castellano; Andrea Esposito; Eufemia Lella; Graziano Montanaro; Gennaro Vessio; et al.

Journal: IEEE Access

Abstract:

Recent advances in deep learning and imaging technologies have revolutionized automated medical image analysis, especially in diagnosing Alzheimer's disease through neuroimaging. Despite the availability of various imaging modalities for the same patient, the development of multi-modal models leveraging these modalities remains underexplored. This paper addresses this gap by proposing and evaluating classification models using 2D and 3D MRI images and amyloid PET scans in uni-modal and multi-modal frameworks. Our findings demonstrate that models using volumetric data learn more effective representations than those using only 2D images. Furthermore, integrating multiple modalities enhances model performance over single-modality approaches significantly. We achieved state-of-the-art performance on t

he OASIS-3 cohort. Additionally, explainability analyses with Grad-CAM indicate that our model focuses on crucial AD-related regions for its predictions, underscoring its potential to aid in understanding the disease's causes.

Paper 49:

Disentangle First, Then Distill: A Unified Framework for Missing Modality Imputation and Alzheimer's Disease Diagnosis (2023)

Author:Yuanyuan Chen; Yongsheng Pan; Yong Xia; Yixuan Yuan. (2023).

Journal: IEEE Access

Abstract:

Multi-modality medical data provide complementary information, and hence have been widely explored for computer-aided AD diagnosis. However, the research is hindered by the unavoidable missing-data problem, i.e., one data modality was not acquired on some subjects due to various reasons. Although the missing data can be imputed using generative models, the imputation process may introduce unrealistic information to the classification process, leading to poor performance. In this paper, we propose the Disentangle First, Then Distill (DFTD) framework for AD diagnosis using incomplete multi-modality medical images. First, we design a region-aware disentanglement module to disentangle each image into inter-modality relevant representation and intra-modality specific representation with emphasis on disease-related regions. To progressively integrate multi-modality knowledge, we then construct an imputation-induced distillation module, in which a lateral inter-modality transition unit is created to impute representation of the missing modality. The proposed DFTD framework has been evaluated against six existing methods on an ADNI dataset with 1248 subjects. The results show that our method has superior performance in both AD-CN classification and MCI-to-AD prediction tasks, substantially over-performing all competing methods

Paper 50:

Leveraging Brain MRI for Biomedical Alzheimer's Disease Diagnosis Using Enhanced Manta Ray Foraging Optimization Based Deep Learning (2023)

Author: R. Syed Jamalullah; L. Mary Gladence; Mohammed Altaf Ahmed; E. Laxmi Lydia; Mohamad Khairi Ishak; Myriam Hadjouni; Samih M. Mostafa

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is the most frequent method of dementia and ranks the fifth-leading disease. Brain imaging or neuroimaging like magnetic resonance imaging (MRI) was utilized in the medicinal analysis of brain condition to allow visualization of infrastructure and brain functionality. Machine learning (ML) approaches on MRI are utilized in the analysis of AD to accelerate the analysis procedure and assist doctors. But, in typical ML approaches utilizing handcrafted extracting feature systems on MRI are difficult, and requires the contribution of expert users. So, executing deep learning (DL) as an automatic extracting feature system is minimizing require for extracting features and automate the procedure. This article introduces a novel Biomedical Alzheimer's Disease Diagnosis using Enhanced Manta ray Foraging Optimization based Deep Learning (ADD-EMRFODL) technique on brain MRI. The presented ADD-EMRFODL technique exploits Gabor filtering (GF) technique as a pre-processing step. Also, the presented ADD-EMRFODL technique utilizes densely connected network (DenseNet-121) model to derive feature vectors. Finally, the ADD process is performed via EMRFO with back propagation neural network (BPNN) classifier. The parameter tuning of the BPNN occurs utilizing the EMRFO algorithm with the intention of enhanced classification accuracy. To depict the improvised ADD performance of the ADD-EMRFODL methodology, a comprehensive set of simulations was effectuated. The results stated the improved outcomes of the ADD-EMRFODL algorithm over other current methodologies.

Kesavan Sithan

LITERATURE REVIEW

Paper 51:

Federated Explainable AI-Based Alzheimer's Disease Prediction With Multimodal Data (2025)

Author: Sobhana Jahan; Md. Rawnak Saif Adib; Syed Mahmudul Huda; Md. Sazzadur Rahman; M. Shamim Kaiser; A. S. M. Sanwar Hosen; Deepak Ghimire; Mi Jin Park

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is a progressive neurological disease that severely impairs cognitive function. Early detection is critical for effective treatment and management. Machine Learning (ML) methods are often used to ensure early detection and prediction. However, ML has various issues, including the data island problem. The fragmentation that results from the data island problem makes building reliable, effective ML models more complex, and it is particularly problematic in industries where privacy is a concern, like healthcare. Federated Learning (FL) can help tackle the data island problem by keeping sensitive patient data decentralized and enabling many institutions to work together on model training without exchanging raw data, all while maintaining privacy compliance. As Random Forest (RF) is proven to be the best-performing classifier in this research, an RF classifier is used to create FL. The model incorporates multiple data modalities, such as Magnetic Resonance Imaging (MRI) segmentation and clinical and psychological data, to capture the variety of characteristics influencing the progress of AD. Another concerning issue with ML is its uninterpretable character. We use SHapley Additive exPlanations (SHAP) Explainable Artificial Intelligence (XAI) techniques that emphasize important factors impacting model decisions in order to improve predictability and transparency. This explainability promotes confidence in AI-based diagnoses by enabling researchers and physicians to comprehend the underlying mechanisms guiding the predictions. The combination of XAI, FL, and Open Access Series of Imaging Studies (OASIS-3) Multimodal data offers an interpretable, scalable, reliable, and privacy-centered solution for multiple complex issues, such as predicting AD. This approach results in better diagnosis precision, greater security, and increased confidence in AI technologies, making it a novel methodology in medical sciences. With data privacy maintained, our metho...

Paper 52:

A Novel Deep Learning Architecture Optimization for Multiclass Classification of Alzheimer's Disease Level (2024).

Author: Mahir Kaya; Yasemin Çetin-Kaya

Journal: IEEE Access

Abstract:

Alzheimer's disease is a neurodegenerative disorder prevalent in older adults, and early diagnosis is crucial for effective treatment. A deep learning model can automatically classify Alzheimer's disease from magnetic resonance imaging to aid clinicians in diagnosis. Convolutional Neural Networks (CNNs) are commonly used for disease detection in medical images, but their performance is limited due to inadequate labeled data, high inter-class similarity, and overfitting problems. Key hyperparameters influencing CNN performance include the number of convolution layers and filters assigned to each convolution layer. About other hyperparameters, numerous combinations exist. Since CNN models take a long time to train, it is quite costly to try all combinations to find the optimal model. Existing studies have optimized only a few hyperparameters, such as learning rate, batch size, and optimizer in custom and transfer learning models. In this study, we propose an algorithm based on particle swarm optimization to fine-tune the hyperparameters, including the number of convolution layers, filters, and other hyperparameters, in CNN architectures designed to classify Alzheimer's disease severity. Using the proposed lightweight model, Alzheimer's disease was accurately classified with an accuracy of 99.53% and an F1-score of 99.63% on a public dataset. Our model surpasses the performance of previous studies, offering the potential to alleviate the burden on doctors and expedite their decision-making processes. The developed framework can be accessed via the link: "<https://ai.gop.edu.tr/alzheimer>".

Paper 53:

A Novel Image Casting and Fusion for Identifying Individuals at Risk of Alzheimer's Disease Using MRI and PET Imaging (2024)

Author: Adediji A. Fakoya; Simon Parkinson

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a neurodegenerative condition characterised by permanent cognitive decline. The financial burden associated with providing care for those affected by this ailment has escalated to a significant magnitude, amounting to billions of dollars. Moreover,

projections indicate that the prevalence of this sickness is anticipated to surge by more than 200% throughout the next 15-year period. Numerous researchers have dedicated their efforts to the field of computer-aided diagnosis (CAD), whereby computational methods are used to discern and diagnose various medical conditions. Nevertheless, the identification of persons afflicted with the condition does not mitigate the adverse consequences. It would be advantageous to identify individuals who may be susceptible to the onset of AD at an early stage. This might enhance the efficacy of pharmaceutical interventions and the implementation of preventive measures for AD. The challenge is to determine the best method for predicting the conversion from mild cognitive impairment (MCI) to AD while accounting for two factors: (a) how to represent the three-dimensional (3D) magnetic resonance imaging (MRI) and positron emission tomography (PET) images, since processing these images takes a long time, a lot of computational power, and the accuracy may not be very good; and (b) how best to combine the images while maintaining their individual image information. To solve the problem, this research examined two imaging modalities, namely MRI and PET, and convolutional neural networks (CNNs). The modalities were used to investigate the creation of a method for selecting slices from 3D scans for eventual conversion into 2D images and to combine the imagery obtained from the slice selection of the modalities into one used in the CNNs. The purpose of developing the integration was to tackle the issue of insufficient data and retain the information from the two modalities while improving the accuracy in performance sim...

Paper 54

Prediction of Alzheimer's Disease Progression With Multi-Information Generative Adversarial Network(2021)

Author:Yan Zhao; Baoqiang Ma; Pengbo Jiang; Debin Zeng; et al.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a chronic neurodegenerative disease, and its long-term progression prediction is definitely important. The structural Magnetic Resonance Imaging (sMRI) can be used to characterize the cortical atrophy that is closely coupled with clinical symptoms in AD and its prodromal stages. Many existing methods have focused on predicting the cognitive scores at future time-points using a set of morphological features derived from sMRI. The 3D sMRI can provide more massive information than the cognitive scores. However, very few works consider to predict an individual brain MRI image at future time-points. In this article, we propose a disease progression prediction framework that comprises a 3D multi-information generative adversarial network (mi-GAN) to predict what one's whole brain will look like with an interval, and a 3D DenseNet based multi-class classification network optimized with a focal loss to determine the clinical stage of the estimated brain. The mi-GAN can generate high-quality

individual 3D brain MRI image conditioning on the individual 3D brain sMRI and multi-information at the baseline time-point. Experiments are implemented on the Alzheimer's Disease Neuroimaging Initiative (ADNI). Our mi-GAN shows the state-of-the-art performance with the structural similarity index (SSIM) of 0.943 between the real MRI images at the fourth year and the generated ones. With mi-GAN and focal loss, the pMCI vs. sMCI accuracy achieves 6.04% improvement in comparison with conditional GAN and cross entropy loss.

Paper:55

A Novel AI-Based System for Detection and Severity Prediction of Dementia Using MRI (2021)

Author:Varun Jain; Om Nankar; Daryl Jacob Jerrish; Shilpa Gite; et al.

Journal: IEEE Access

Abstract:

Dementia is a symptom of Alzheimer's Disease (A.D.) that affects many people around the globe each year. There is no effective cure to treat this disease, and it can prove to be deadly to the patient if left untreated or undetected. In this paper, the authors propose a novel DCGAN-based Augmentation and Classification (D-BAC) model approach to identify and classify dementia into various categories depending upon its prominence and severity in the available MRI scans. The proposed detection of early onset of dementia, also referred to as Mild Cognitive Impairment (MCI), is also studied with the help of a novel GAN-augmented dataset. The proposed model can predict MCI with an accuracy of 74% and can classify dementia into four categories depending upon its prominence in the MRI scan. The authors have also utilized Visual Explainable A.I. (XAI) and have used GradCAM to visually represent the internal working of the model. This novel approach helps verify the differentiating features of the MRI scans learned by the CNN model during training. Three different datasets, namely the original dataset, geometrically transformed images, and a GAN-augmented dataset, have been used for performance analysis. A comparison of the performance of the CNN model has been made on these datasets, and the superiority of results using the novel GAN-augmented dataset has been studied and discussed. Moreover, progressive resizing has also been applied on this GAN-dataset, and different CNN architectures have also been used to achieve better performance scores. The model proposed in the end has a training accuracy of 97% and a testing accuracy of 82% when tested using a conventional CNN architecture and has a testing accuracy of 84% and 87% when tested using VGG-16 and VGG-19 architecture, respectively.

Paper 56

Dual Attention Multi-Instance Deep Learning for Alzheimer's Disease Diagnosis With Structural MRI (2021)

Author:Wenyong Zhu; Liang Sun; Jiashuang Huang; Liangxiu Han; Daoqiang Zhang.

Journal: IEEE Access

Abstract:

structural magnetic resonance imaging (sMRI) is widely used for the brain neurological disease diagnosis, which could reflect the variations of brain. However, due to the local brain atrophy, only a few regions in sMRI scans have obvious structural changes, which are highly correlative with pathological features. Hence, the key challenge of sMRI-based brain disease diagnosis is to enhance the identification of discriminative features. To address this issue, we propose a dual attention multi-instance deep learning network (DA-MIDL) for the early diagnosis of Alzheimer's disease (AD) and its prodromal stage mild cognitive impairment (MCI). Specifically, DA-MIDL consists of three primary components: 1) the Patch-Nets with spatial attention blocks for extracting discriminative features within each sMRI patch whilst enhancing the features of abnormally changed micro-structures in the cerebrum, 2) an attention multi-instance learning (MIL) pooling operation for balancing the relative contribution of each patch and yield a global different weighted representation for the whole brain structure, and 3) an attention-aware global classifier for further learning the integral features and making the AD-related classification decisions. Our proposed DA-MIDL model is evaluated on the baseline sMRI scans of 1689 subjects from two independent datasets (i.e., ADNI and AIBL). The experimental results show that our DA-MIDL model can identify discriminative pathological locations and achieve better classification performance in terms of accuracy and generalizability, compared with several state-of-the-art methods.

Paper 57:

Biceph-Net: A Robust and Lightweight Framework for the Diagnosis of Alzheimer's Disease Using 2D-MRI Scans and Deep Similarity Learning

Author:A. H. Rashid; Aditya Gupta; Jhalak Gupta; M. Tanveer.

Journal: IEEE Access

Abstract

Alzheimer's Disease (AD) is a neurodegenerative disease that is one of the significant causes of death in the elderly population. Many deep learning techniques have been proposed to diagnose AD using Magnetic Resonance Imaging (MRI) scans. Predicting AD using 2D slices extracted from 3D MRI scans is challenging as the inter-slice information gets lost. To this end,

we propose a novel and lightweight framework termed 'Biceph-net' for AD diagnosis using 2D MRI scans that models both the intra-slice and inter-slice information. 'Biceph-net' has been experimentally shown to perform similar to other Spatio-temporal neural networks while being computationally more efficient. Biceph-net is also superior in performance compared to vanilla 2D convolutional neural networks (CNN) for AD diagnosis using 2D MRI slices. Biceph-net also has an inbuilt neighbourhood-based model interpretation feature that can be exploited to understand the classification decision taken by the network. Biceph-net experimentally achieves a test accuracy of 100% in the classification of Cognitively Normal (CN) vs AD, 98.16% for Mild Cognitive Impairment (MCI) vs AD, and 97.80% for CN vs MCI vs AD.

Paper 58:

ADD-Net: An Effective Deep Learning Model for Early Detection of Alzheimer Disease in MRI Scans (2022).

Author: Muhammad M. S. Fareed; S. Zikria; G. Ahmed; Mui-zzud-din; S. Mahmood; M. Aslam; S. F. Jilani; A. Moustafa; M. Asad.

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is a neurological brain disorder marked by dementia and neurological dysfunction that affects memory, behavioral patterns, and reasoning. Alzheimer's disease is an incurable disease that primarily affects people over 40. Alzheimer's disease is diagnosed through a manual evaluation of a patient's MRI scan and neuro-psychological examinations. Deep Learning (DL), a type of Artificial Intelligence (AI), has pioneered new approaches to automate medical image diagnosis. This study aims to create a reliable and efficient system for classifying AD using MRI by applying the deep Convolutional Neural Network (CNN). In this paper, we propose a new CNN architecture for detecting AD with relatively few parameters, and the proposed solution is ideal for training a smaller dataset. This proposed model successfully distinguishes the early stages of Alzheimer's disease and shows class activation maps as a heat map on the brain. The proposed Alzheimer's Disease Detection Network (ADD-Net) is built from scratch to precisely classify the stages of AD by decreasing parameters and calculation costs. The Kaggle MRI image dataset has a significant class imbalance problem, and we exploited a synthetic oversampling technique to evenly distribute the image among the classes to prevent the problem of class imbalance. The proposed ADD-Net is extensively evaluated against DenseNet169, VGG19, and InceptionResNet V2 using precision, recall, F1-score, Area Under the Curve (AUC), and loss. The ADD-Net achieved the following values for evaluation metrics: 98.63%, 99.76%, 98.61%, 98.63%, 98.58%, and 0.0549% accuracy, AUC, F1-score, precision, recall, and loss, respectively. The simulation results show that the proposed ADD-Net outperforms other state-of-the-art models in all the evaluation metrics

Paper 59

Integrating Multimodal AI Techniques and MRI Preprocessing for Enhanced Diagnosis of Alzheimer's Disease: Clinical Applications and Research Horizons (2025)

Author:Bandaru A. Chakravarthi; Gandla Shivakanth

Journal: IEEE Access

Abstract:

A progressive neurological disorder, Alzheimer's disease has a significant social and economic impact and has become more problematic for world health. Early and precise diagnosis is critical to enable timely intervention to improve patient outcomes. This study introduces a new multimodal AI framework integrating advanced MRI preprocessing techniques and speech analysis for the enhancement of the detection of AD. A strong preprocessing pipeline, including noise reduction, normalization, skull stripping, and segmentation, is utilized to prepare the MRI data. Cognitive and acoustic biomarkers are extracted through spectrograms and pre-trained linguistic models to create speech features. The architecture involves the application of Vision Transformers for spatial analysis and a hybrid CNN- RNN architecture to derive contextual insights, making the framework applicable to multimodal fusion for holistic diagnosis. The framework achieved an accuracy of 94.2% with precision, recall, and F1-scores higher than 92% in the experimental evaluation on a diverse dataset. Comparative analysis with several recent studies further emphasizes that the framework provides better diagnostic performance on key metrics. This work advances early AD detection by incorporating complementary data modalities, addressing existing gaps in unimodal approaches, and providing a scalable, interpretable solution for clinical application.

Paper 60:

DeepCurvMRI: Deep Convolutional Curvelet Transform-Based MRI Approach for Early Detection of Alzheimer's Disease (2023)

Author: Chahd Chabib; Leontios J. Hadjileontiadis; Aamna A. Al Shehhi.

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) is the most common form of dementia. It usually manifests through progressive loss of cognitive function and memory, subsequently impairing the person's ability to live without assistance and causing a tremendous impact on the affected individuals and society. Currently, AD diagnosis relies on cognitive tests, blood tests, behavior assessments, brain imaging, and medical history analysis. However, these procedures are subjective and inconsistent, making an accurate prediction for the early stages of AD difficult. This paper introduces a curvelet transform (CT) based-convolutional neural network (CNN) (DeepCurvMRI)

model for improving the accuracy of early-stage AD disease detection using from Magnetic resonance imaging (MRI) images. The MRI images were first pre-processed using CT, and then a CNN model was trained using the new image representation. In this study, we used Alzheimer's MRI images dataset hosted on the Kaggle platform to train DeepCurvMRI for multi and binary classification tasks. DeepCurvMRI achieved an accuracy, sensitivity, specificity, and F1 score of $98.62\% \pm 0.10\%$, $99.05\% \pm 0.10\%$, $98.50\% \pm 0.03\%$, and 99.21 ± 0.08 , respectively, using the leave-one-group-out (LOGO) cross-validation approach in multi-classification task. The highest accuracy obtained in binary classification is $98.71\% \pm 0.05\%$. In addition to LOGO, DeepCurvMRI was tested using randomly stratified 10-fold and 5-fold cross validation. These encouraging results are superior to the ones reported in related methods, showcasing the potentiality of DeepCurvMRI in capturing the key anatomical changes in MRI images that can be differentiated between various staged of Alzheimer's disease classes.

Paper 61

A FixMatch Framework for Alzheimer's Disease Classification: Exploring the Trade-Off Between Supervision and Performance (2025)

Author: Al Hossain; Umme Hani Konok; Md Tahsin; Raihan UI Islam; Mohammad Rifat Ahmmad Rashid; Mohammad Shahadat Hossain; Karl Andersson

Journal: IEEE Access

Abstract:

Alzheimer's Disease (AD) poses a major challenge for healthcare systems worldwide, as timely and accurate diagnosis is crucial for patient management and outcome improvement. While experienced medical professionals can often identify AD through conventional assessment methods, limited resources and growing patient populations make large-scale and rapid screening increasingly necessary. In this work, we explore whether the FixMatch algorithm—a semi-supervised learning approach—can aid in classifying Alzheimer's Disease (AD), Mild Cognitive Impairment (MCI), and Cognitively Normal (CN) by using the ADNI fMRI dataset of 5,182 images. This approach supplements rather than replaces clinical expertise, offering a faster, more standardized classification process where expert labeling is limited. We first assessed various supervised models and determined that VGG19-FFT provided the strongest balance of classification accuracy and computational efficiency. Integrating VGG19-FFT into FixMatch as the teacher model, initial tests using 10% labeled and 90% unlabeled data yielded modest results. A more systematic examination of different labeled-to-unlabeled data splits revealed that a 60:40 ratio enabled FixMatch to achieve classification accuracies of 100% for AD, 99% for CN, and 99% for MCI—on par with fully supervised training. This outcome highlights the potential of FixMatch to significantly reduce labeling requirements, a particular advantage in resource-constrained settings where expert annotations are costly. By striking an effective balance between labeling effort and model performance, the identified 60:40 ratio helps

make advanced diagnostic methods both feasible and practical in real-world clinical applications.

Paper 62

An Improved Deep Learning Unsupervised Approach for MRI Tissue Segmentation for Alzheimer's Disease Detection (2024)

Author: Karan Kumar; Isha Suwalka; Adaora Uche-Ezennia; Celestine Iwendi; Cresantus N. Biamba

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) ranks as the sixth leading cause of death, emphasizing the need for early-stage prediction to prevent its progression. Due to the complexity and heterogeneity of medical tests, manually comparing, visualizing, and analyzing data is often difficult and time-consuming. As a result, a computational approach for accurately predicting brain changes through the classification of magnetic resonance imaging (MRI) scans becomes highly valuable, though challenging. This paper introduces a novel method for diagnosing the early stages of AD by utilizing an efficient mapping technique to differentiate between affected and normal MRI scans. The approach combines a hybrid unsupervised learning framework, specifically the adaptive moving self-organizing map (AMSOM) method integrated with Fuzzy K-means. To ensure optimal feature extraction, we introduce a hybrid learning framework that embeds feature vectors in a subspace. The analysis compares various mapping approaches to identify features linked to Alzheimer's disease. The proposed method achieves a classification accuracy of 95.75% on the Open Access Series of Imaging Studies (OASIS) MRI brain image database, outperforming existing methods.

Paper 63:

Adversarial Network-Based Classification for Alzheimer's Disease Using Multimodal Brain Images: A Critical Analysis (2024).

Author: Meenu Gupta; Rakesh Kumar; Ajith Abraham.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that represents a significant and growing public health challenge. This work concisely summarizes AD, encompassing its pathophysiology, risk factors, clinical manifestations, diagnosis, treatment,

and ongoing research. The main goal of managing AD is to reduce symptoms while improving the lives of those impacted. This letter has conducted a systematic review to analyze the prediction of AD using the Preferred Reporting Item for Systematic Review and Meta-Analysis (PRISMA) guidelines. The major scientific databases such as Scopus, Web of Science (WoS), and IEEE Xplorer are explored, where 2018–2023 publications are considered. The article selection process is based on keywords like “Alzheimer’s disease,” “Brain Images,” “Deep Learning (DL),” etc. After rigorous analysis, 946 articles were extracted, and 42 were identified for final consideration. Further, several investigations based on the previous work are discussed along with its Proposed Solutions (PS). Finally, a case study on AD detection using the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset and AD Detection Network (ADD-NET) implementation is presented.

Paper 64:

Deep Learning Applications in MRI-Based Detection of the Hippocampal Region for Alzheimer’s Diagnosis (2024)

Author: Yori Pusparani; Chih-Yang Lin; Yih-Kuen Jan; Fu-Yu Lin; Ben-Yi Liao; Peter Ardhianto; Elvin Nur Furqon; John Sahaya Rani Alex; Jeetashree Aparajeeta; Chi-Wen Lung

Journal: IEEE Access

Abstract:

The hippocampal region is one of the most affected brain areas observed as a landmark in Magnetic Resonance Imaging (MRI) images for Alzheimer’s disease (AD) diagnosis. The diminished alterations in the hippocampal and degeneration of cholinergic circuits have been conclusively correlated with a decline in memory and cognitive function. However, the hippocampal region may not appear as clearly defined as other brain regions, making it difficult for neurologists and researchers to identify by visual inspection. The application of deep learning models to pinpoint the hippocampal region was initially valued. We assessed the ability of a deep learning model, You Only Live Once (YOLO), to detect hippocampal regions in three MRI image views and categories. The Alzheimer’s Disease Neuroimaging Initiative-first (ADNI-1) dataset was used with 220 subjects in three categories using the three YOLO models. We obtained the YOLO performance for hippocampal region detection with accuracy in three views and categories. The average mean Average Precision (mAP) performance accuracy for YOLOv3 was 0.87, YOLOv4 was 0.85, and YOLOv5 was 0.96, respectively. The high accuracy of the detection of the hippocampal region was remarkable. We found that the sagittal view was higher than the axial and coronal views. Simultaneously, the Mild Cognitive Impairment (MCI) in the coronal view was lower among the three models. The results showed that YOLOv5 is a suitable model for detecting the hippocampal region in MRI images, and the sagittal view is the most reliable for detecting the hippocampal region in diagnosing AD. Our findings demonstrate

the importance of detecting the hippocampal region to diagnose AD and accurately analyzing the hippocampal area within the region. The YOLOv5 model substantially affected performance metrics and interpretability across the three views and categories.

Paper 65:

Disease-Oriented Image Embedding With Pseudo-Scanner Standardization for Content-Based Image Retrieval on 3D Brain MRI (2021)

Author: Hayato Arai; Yuto Onga; Kumpei Ikuta; Yusuke Chayama; Hitoshi Iyatomi; Kenichi Oishi

Journal: IEEE Access

Abstract:

To build a robust and practical content-based image retrieval (CBIR) system applicable to clinical brain MRI databases, we propose a new framework, disease-oriented image embedding with pseudo-scanner standardization (DI-PSS). It consists of two core techniques: data harmonization to absorb differences caused by different scanning environments and an algorithm to generate low-dimensional embeddings suitable for disease classification. Until now, there have been very few studies aimed at CBIR of brain MRI. Even in the harmonization of scanners, which is an important prerequisite technique for CBIR, only a limited number of studies have been conducted on T1-weighted MRI, which has collected a vast amount of clinical data. Recently proposed methods need to correctly estimate the domain (i.e., dataset, scanner) of each data in advance to remove environment-dependent information from low-dimensional embedding, which is not an easy task. With DI-PSS, each brain image is pseudo-transformed into a brain image taken with a given reference scanner. Then, 3D convolutional autoencoders (3D-CAE) trained with deep metric learning generate low-dimensional embeddings that better reflect the characteristics of the disease. In this study, DI-PSS reduced the variability of distance in low-dimensional embedding between Alzheimer's disease (AD) and clinically normal (CN) patients, caused by differences in scanners and datasets, by 15.8-22.6% and 18.0-29.9%, respectively, compared to the baseline. This improved the ability of spectral clustering to classify AD and CN by 6.2% in average accuracy and 10.7% in macro-F1. Our method has the advantage of not requiring difficult domain prediction tasks in advance, and can effectively utilize the big data of T1-weighted MR images. Given the potential of the DI-PSS for harmonizing images scanned by MRI scanners that were not used to scan the training data, it is well suited for application to a large number of legacy MRIs captured in heterogeneous..

Paper 66:

Deep Learning-Based Framework for Predicting Mild Cognitive Impairment Progression in Neurology Using Longitudinal MRI (2025)

Author: Guojiang Zheng; Yang Lu; Hui Chen

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD), a leading neurodegenerative disorder, progresses from an intermediary stage known as Mild Cognitive Impairment (MCI), characterized by measurable cognitive decline with retained functional independence. Accurate prediction of MCI progression to AD is critical for timely interventions. Existing deep learning-based methods for structural MRI (sMRI) analysis predominantly utilize either Convolutional Neural Networks (CNNs), which effectively capture local features but neglect global context, or Transformer architectures that model global dependencies yet require extensive data and computational resources. Additionally, many methods inadequately leverage longitudinal imaging data, limiting their sensitivity to subtle temporal changes in brain morphology. To overcome these limitations, we introduce EffiSwin-MCI, a novel hybrid deep learning framework integrating EfficientNet and Swin Transformer architectures, specifically designed for longitudinal sMRI analysis. The primary novelty of EffiSwin-MCI lies in its sliding-window attention mechanism, inspired by the Swin Transformer, which effectively integrates localized spatial dependencies within 2D sMRI slices, combined with temporal attention blocks that fuse spatial-temporal features across longitudinal scans at two distinct time points (T1 and T2). EfficientNet-B2 serves as a computationally efficient backbone, extracting hierarchical spatial features crucial for detailed morphological characterization. This alternating spatial and temporal attention strategy uniquely captures progressive local and global structural changes indicative of cognitive decline. Comprehensive experiments conducted on the Alzheimer's Disease ADNI dataset demonstrate the proposed model's superior performance compared to state-of-the-art CNN and Transformer-based approaches, achieving an accuracy of 81.69%, recall of 80.27%, precision of 84.35%, and F1-score of 82.27%. EffiSwin-MCI's interpretability is further validated.

Paper 67:

Gradual Variation-Based Dual-Stream Deep Learning for Spatial Feature Enhancement With Dimensionality Reduction in Early Alzheimer's Disease Detection (2025).

Author: Najmul Hassan; Abu Saleh Musa Miah; Taro Suzuki; Jungpil Shin.

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is one of the most common neurological brain disorders, developing gradually and primarily causing the deterioration of brain function, leading to memory loss and behavioral changes. Early detection through computer-aided systems is crucial for effective intervention and treatment, providing automated and highly accurate diagnoses of Alzheimer's

by analyzing brain scans, such as MRI, to assess atrophy in key regions like the hippocampus and entorhinal cortex. Recently, many researchers have been developing computer-aided AD recognition systems using various technologies. However, existing approaches face challenges in achieving good performance due to difficulties in capturing complex spatial features and handling high-dimensional data. This study proposes a Gradual Variation-Based Dual-Stream Deep Learning (DL) framework for Spatial Feature Enhancement with Dimensionality Reduction to address these challenges. The method begins by converting MRI voxel data into slice images, followed by the extraction of gradual spatial variations based on the Haar Wavelet Transform (WT) to capture large-scale structural changes. These features are processed through a dual-stream DL model, where each stream captures complementary spatial features. The concatenated features are refined using a channel attention (CA) module, which selects the most discriminative features while performing dimensionality reduction to retain only the most important information, eliminating redundant or less informative features. Finally, a DL-based classification module is applied to detect early signs of Alzheimer's disease. We evaluated the model on two datasets, Complete 1Yr 1.5T and Complete 3Yr 3T, from the Alzheimer's Disease Neuroimaging Initiative (ADNI), using MRI scans from three planes (axial, sagittal, and coronal). The proposed method achieved 99.05% accuracy, 99.05% precision, 99.05% recall, 99.73% ROC-AUC, and 99.33% F1-score, significantly outperforming state-of-the-art methods. These results demonstrate the robustness and computational efficiency of our approach for early Alzheimer's diagnosis, offering a valuable tool for clinical applications

Paper 68:

Attention-Enhanced Dual-Path CNN for Early Alzheimer's Detection From Multi-Planar T1-Weighted MRI (2025)

Author: Vyshnavi Ramineni; Jun-Hyung Kim; Chun-Su Park; Ji-In Kim; Goo-Rak Kwon. (2025).

Journal: IEEE Access

Abstract:

Alzheimer's disease (AD) is a progressive neurodegenerative disorder, and early detection is essential for effective clinical intervention. Previous deep learning approaches have achieved promising results using MRI data; however, most rely on full 3D volumes or standard 2D axial and coronal slices, often overlooking the diagnostic value of parasagittal views. In addition, many existing models lack attention mechanisms and multi-branch architectures, limiting their ability to capture both localized and contextual features. To overcome these limitations, we propose an attention-guided dual-path CNN that integrates sagittal, coronal, and parasagittal slices extracted at a 6.257° off-midline angle. The architecture combines a focused SNeurodCNN branch with an Inception-v4 path enhanced by CBAM for multi-scale feature extraction. Using T1-weighted MRI scans from the ADNI database, the proposed model

achieved an accuracy of 98.9% and an AUC of 0.992, highlighting its potential for accurate and early AD classification.

Paper 69:

Adversarial Learning for MRI Reconstruction and Classification of Cognitively Impaired Individuals (2023).

Author:Xiao Zhou; Akshara R. Balachandra; Michael F. Romano; Sang Peter Chin;

Journal: IEEE Access

Abstract:

Game theory-inspired deep learning using a generative adversarial network provides an environment to competitively interact and accomplish a goal. In the context of medical imaging, most work has focused on achieving single tasks such as improving image resolution, segmenting images, and correcting motion artifacts. We developed a dual-objective adversarial learning framework that simultaneously 1) reconstructs higher quality brain magnetic resonance images (MRIs) that 2) retain disease-specific imaging features critical for predicting progression from mild cognitive impairment (MCI) to Alzheimer's disease (AD). We obtained 3-Tesla, T1-weighted brain MRIs of participants from the Alzheimer's Disease Neuroimaging Initiative (ADNI, N=342) and the National Alzheimer's Coordinating Center (NACC, N = 190) datasets. We simulated MRIs with missing data by removing 50% of sagittal slices from the original scans (i.e., diced scans). The generator was trained to reconstruct brain MRIs using the diced scans as input. We introduced a classifier into the GAN architecture to discriminate between stable (i.e., sMCI) and progressive MCI (i.e., pMCI) based on the generated images to facilitate encoding of disease-related information during reconstruction. The framework was trained using ADNI data and externally validated on NACC data. In the NACC cohort, generated images had better image quality than the diced scans (Structural similarity (SSIM) index: 0.553 ± 0.116 versus 0.348 ± 0.108). Furthermore, a classifier utilizing the generated images distinguished pMCI from sMCI more accurately than with the diced scans (F1-score: 0.634 ± 0.019 versus 0.573 ± 0.028). Competitive deep learning has potential to facilitate disease-oriented image reconstruction in those at risk of developing Alzheimer's disease.

Paper 70:

Multi-Modal Integration Analysis of Alzheimer's Disease Using Large Language Models and Knowledge Graphs (2025)

Author:Kanan Kiguchi; Yunhao Tu; Katsuhiko Ajito; Fady Alnajjar; Kazuyuki Murase

Journal: IEEE Access

Abstract:

We propose a novel framework for integrating fragmented multi-modal data in Alzheimer's disease (AD) research using large language models (LLMs) and knowledge graphs. While traditional multi-modal analysis requires matched patient IDs across datasets, our approach demonstrates population-level integration of MRI, gene expression, biomarkers, EEG, and clinical indicators from independent cohorts. Statistical analysis identified significant features in each modality, which were connected as nodes in a knowledge graph. LLMs then analyzed the graph to extract potential correlations and generate hypotheses in natural language. This approach revealed several novel relationships, including a potential pathway linking metabolic risk factors to tau protein abnormalities via neuroinflammation ($r > 0.6$, $p < 0.001$), and unexpected correlations between frontal EEG channels and specific gene expression profiles ($r = 0.42-0.58$, $p < 0.01$). Cross-validation with independent datasets confirmed the robustness of major findings, with consistent effect sizes across cohorts (variance $< 15\%$). The reproducibility of these findings was further supported by expert review (Cohen's $\kappa = 0.82$) and computational validation. Our framework enables cross-modal integration at a conceptual level without requiring patient ID matching, offering new possibilities for understanding AD pathology through fragmented data reuse and generating testable hypotheses for future research.

Paper 71:

Functional Brain Network Measures for Alzheimer's Disease Classification (2023)

Author: Luyun Wang; Jinhua Sheng; Qiao Zhang; Rougang Zhou; Yu Xin; Bo Wang; et al.

Journal: IEEE Access

Abstract:

Background: Alzheimer's disease (AD) is an incurable neurodegenerative disease primarily affecting the elderly population. The therapy of AD depends heavily on an early diagnosis. In this study, our primary objective is to evaluate the classification framework, which combines graph theory and machine learning techniques for functional magnetic resonance imaging (fMRI), to distinguish AD, early mild cognitive impairment (EMCI), late mild cognitive impairment (LMCI), and healthy control (HC). **Methods:** A novel multi-feature selection method, incorporating the dual graph theoretical approach, is proposed for classification. This method utilizes three different feature selection methods after brain areas selection through graph-theory analyses in 96 subjects with brain parcellation by using the joint human connectome project multimodal parcellation (J-HCPMMP) of 180 areas per hemisphere. **Results:** The classification results show that the optimal features selected by the minimal redundancy maximal relevance (MRMR) based on support vector machine linear (SVM-linear) from graph measures for 36 areas of 360 areas. The classification accuracies for identifying HC vs. EMCI, HC vs. LMCI, HC vs. AD, EMCI vs. LMCI, LMCI vs. AD, and EMCI vs. AD, are 85.60%, 92.90%, 96.80%, 83.30%, 84.90% and 89.50%, respectively. **Conclusion:** The results indicate that the combination of graph measures

and machine learning in fMRI connectivity analysis might be helpful for the diagnosis of AD, especially the use of local measures, which may better reflect functional changes in local brain regions because of cognitive impairment.

Paper 72:

CirMNet: A Shape-Based Hybrid Feature Extraction Technique Using CNN and CMSMD for Alzheimer's MRI Classification (2024).

Author: G. Malu; Nayana Uday; Elizabeth Sherly; Ajith Abraham; Narendra Kuber Bodhey.

Journal: IEEE Access

Abstract:

This research introduces a novel approach called Circular Mesh Network (CirMNet), a shape-based hybrid feature extraction technique. It also proposes innovative Fractal Dimension (FD) and statistical feature extraction techniques for vertically symmetrical images. Convolutional Neural Networks (CNNs) have gained widespread popularity across various domains, including the interpretation of medical images. CNNs excel at extracting prominent features in the initial layers and progressively learn to capture more complex features as they advance. However, the pooling and striding operations involved in CNNs can lead to a loss of spatial and structural details in the image because CNNs require a mechanism to preserve the internal representation and describe the intricate relationships between image components or pixels. Circular Mesh-based Shape and Margin Descriptor (CMSMD) focuses on extracting structural, statistical, and property-based features. However, it does not encompass features such as texture or color. The objective of CirMNet is to leverage the strengths of both CNNs and CMSMD, and to mitigate their respective weaknesses. Structural and texture features were generated from CirMNet, and its performance was evaluated for the diagnosis of neurodegenerative disorders, particularly Alzheimer's Disease (AD). The model can easily identify the permanent shrinkage and destruction of brain cells in MRIs of patients and exhibited a notable accuracy of 97.34% in classifying various stages of AD, encompassing Control, Early Mild Cognitive Impairment (EMCI), and Late Mild Cognitive Impairment (LMCI). This achievement represents a substantial improvement over the existing state-of-the-art methods in the domain

Paper 73:

Enhancing Chronic Disease Prediction in IoMT-Enabled Healthcare 5.0 Using Deep Machine Learning: Alzheimer's Disease as a Case Study (2025)

Author: Rabia Javed; Tahir Abbas; Tariq Shahzad; Khadija Kanwal; Sadaqat Ali Ramay; Muhammad Adnan Khan; Khmaies Ouahada.

Journal: IEEE Access

Abstract:

Chronic disease significantly affects health on a global scale. Deep machine learning algorithms have found widespread application in the diagnosis of chronic diseases. Early diagnosis and treatment reduce the chance of a disease getting worse and, as a result, raise related mortality. The main objective of this work is to present a deep machine learning-based approach that provides better results in terms of accuracy. These findings have significance for tailored healthcare 5.0, enabling healthcare professionals to predict chronic disease more efficiently. A comparative examination of the most recent methods has been provided in our work reveals that it might be more advantageous to use the proposed model in which segmentation of the MRI is performed using U-net architecture and then classification is done using transfer learning for chronic disease prediction. Our proposed model provides 96.06% accuracy, it advances our understanding of deep machine learning's potential for chronic disease prediction and emphasizes the need to tailor model selection to specific disease types using data from IoMT enabled devices. In order to make advanced improvement in the field of healthcare 5.0, future studies should focus on refining these models and investigating how well they work with a wider range of datasets

Paper 74:

Leveraging Bi-Focal Perspectives and Granular Feature Integration for Accurate Reliable Early Alzheimer's Detection (2025).

Author Shravan Venkatraman; Pandiyaraju Vairamoorthy; A. Abeshek; S. Pavan Kumar; et al.

Journal: IEEE Access

Abstract:

Being the most commonly known neurodegeneration, Alzheimer's Disease (AD) is annually diagnosed in millions of patients. The present medical scenario still finds the exact diagnosis and classification of AD through neuroimaging data as a challenging task. Traditional CNNs can extract a good amount of low-level information in an image while failing to extract high-level minuscule particles, which is a significant challenge in detecting AD from MRI scans. To overcome this, we propose a novel Granular Feature Integration method to combine information extraction at different scales along with an efficient information flow, enabling the model to capture both broad and fine-grained features simultaneously. We also propose a Bi-Focal Perspective mechanism to highlight the subtle neurofibrillary tangles and amyloid plaques in the MRI scans, ensuring that critical pathological markers are accurately identified. Our model achieved an F1-Score of 99.31%, precision of 99.24%, and recall of 99.51%. These scores prove that our model is significantly better than the state-of-the-art (SOTA) CNNs in existence.

Paper 75:

Alzheimer's Disease and Mild Cognitive Impairment Detection Using sMRI With Efficient Receptive Field and Enhanced Multi-Axis Attention Fusion (2024)

Author: Uttam Khatri; Jun-Hyung Kim; Goo-Rak Kwon.

Journal: IEEE Access

Abstract:

Deep neural networks have shown promising results in the analysis of structural magnetic resonance imaging (sMRI) data for the diagnosis of dementia, particularly Alzheimer's disease (AD). Different regions of the brain have diverse structures that are linked to specific functions, which could account for the variability in disease-related changes observed in sMRI scans of these areas. Understanding the overall characteristics of sMRI data is important since current popular convolutional neural networks (CNN) for deep learning do not consider the interconnection of voxels. Vision transformers have shown effectiveness in identifying long-distance connections in the brain, which has led to their success in applications such as disease detection. However, the image noise and limited scalability of self-attention mechanisms in relation to image size has hindered their widespread use in advanced Alzheimer's analysis. To enhance information retention and reduce network complexity, this study presents a novel adaptable efficient receptive field feature extraction network. Moreover, an advanced attention mechanism with both local grid attention block and dilated global attention module has been incorporated to highlight the characteristics of AD. Next, a more improved hierarchical inverted residual feed forward network in place of multi-layer perceptron is suggested to enhance the characterization of features through the integration of information from both lower and higher layers. Finally, the global average pooling and 1×1 convolution are used to reduce dimensionality, enhance non-linearity, and allow channel interactions in feature maps before being input into the classification head.

